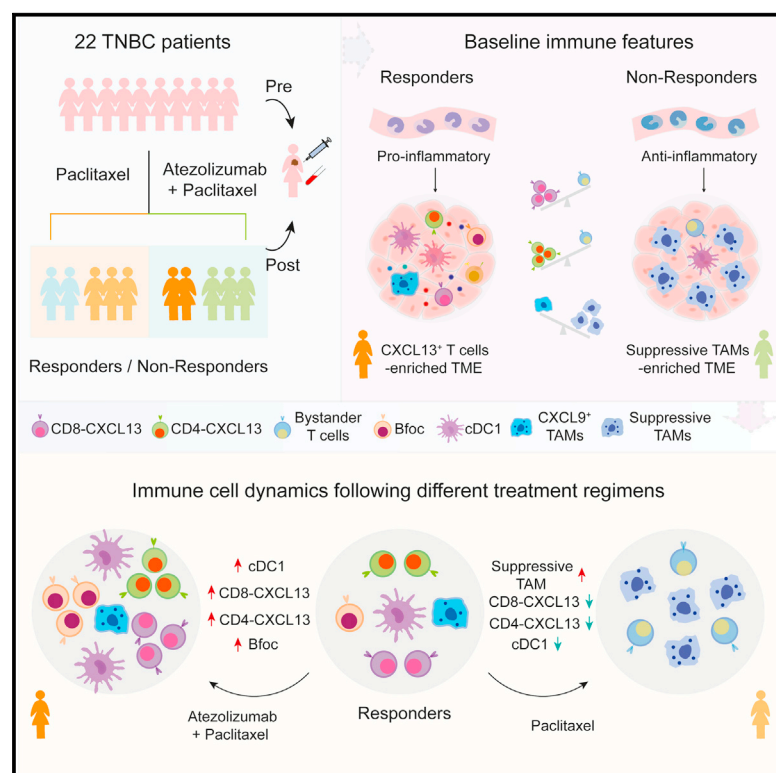


Single-cell analyses reveal key immune cell subsets associated with response to PD-L1 blockade in triple-negative breast cancer

Graphical abstract



Authors

Yuanyuan Zhang, Hongyan Chen, Hongnan Mo, ..., Fei Ma, Zemin Zhang, Zhihua Liu

Correspondence

xubinghe@cscsco.org.cn (B.X.),
drmafei@126.com (F.M.),
zemin@pku.edu.cn (Z.Z.),
liuzh@cicams.ac.cn (Z.L.)

In brief

Zhang et al. combine single-cell RNA-seq, TCR-seq, and ATAC-seq to investigate immune cell dynamics in the tumor microenvironment and peripheral blood of patients with TNBC treated with paclitaxel or paclitaxel plus atezolizumab, revealing immune features of responders and nonresponders, the mechanisms and intertwined effects of paclitaxel and atezolizumab in TNBC treatment.

Highlights

- Dynamic immune cell alterations revealed in TNBC following anti-PD-L1 treatment
- CD8-CXCL13 and CD4-CXCL13 T cells predict effective responses to PD-L1 blockade
- Proinflammatory macrophages expressing CXCL9 are associated with CXCL13⁺ T cells
- Paclitaxel impairs the expansion of responsive immune cells caused by atezolizumab



Article

Single-cell analyses reveal key immune cell subsets associated with response to PD-L1 blockade in triple-negative breast cancer

Yuanyuan Zhang,^{1,7} Hongyan Chen,^{2,7} Hongnan Mo,^{3,7} Xueda Hu,^{1,7} Ranran Gao,¹ Yahui Zhao,² Baolin Liu,¹ Lijuan Niu,⁴ Xiaoying Sun,⁵ Xiao Yu,² Yong Wang,⁴ Qing Chang,⁴ Tongyang Gong,² Xiuwen Guan,³ Ting Hu,² Tianyi Qian,³ Binghe Xu,^{2,3,*} Fei Ma,^{2,3,*} Zemin Zhang,^{1,6,8,*} and Zhihua Liu^{2,*}

¹BIOPIC, Peking-Tsinghua Center for Life Sciences, Academy for Advanced Interdisciplinary Studies, School of Life Sciences, International Cancer Institute, Peking University, Beijing 100871, China

²State Key Laboratory of Molecular Oncology, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing 100021, China

³Department of Medical Oncology, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing 100021, China

⁴Department of Ultrasound, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing 100021, China

⁵Department of Medical Oncology, Cancer Hospital of HuanXing ChaoYang District, Beijing 100005, China

⁶Institute of Cancer Research, Shenzhen Bay Laboratory, Shenzhen 518132, China

⁷These authors contributed equally

⁸Lead contact

*Correspondence: xubinghe@csc.org.cn (B.X.), drmafei@126.com (F.M.), zemin@pku.edu.cn (Z.Z.), liuzh@cicams.ac.cn (Z.L.)

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SUMMARY

In triple-negative breast cancer (TNBC), the benefit of combining chemotherapy with checkpoint inhibitors is still not very clear. We utilize single-cell RNA- and ATAC-sequencing to examine the immune cell dynamics in 22 patients with advanced TNBC treated with paclitaxel or its combination with the anti-PD-L1 atezolizumab. We demonstrate that high levels of baseline CXCL13⁺ T cells are linked to the proinflammatory features of macrophages and can predict effective responses to the combination therapy. In responsive patients, lymphoid tissue inducer (LTi) cells, follicular B (B₁₀) cells, CXCL13⁺ T cells, and conventional type 1 dendritic cells (cDC1) concertedly increase following the combination therapy, but instead decrease after paclitaxel monotherapy. Our data highlight the importance of CXCL13⁺ T cells in effective responses to anti-PD-L1 therapies and suggest that their reduction by paclitaxel regimen may compromise the clinical outcomes of accompanying atezolizumab for TNBC treatment.

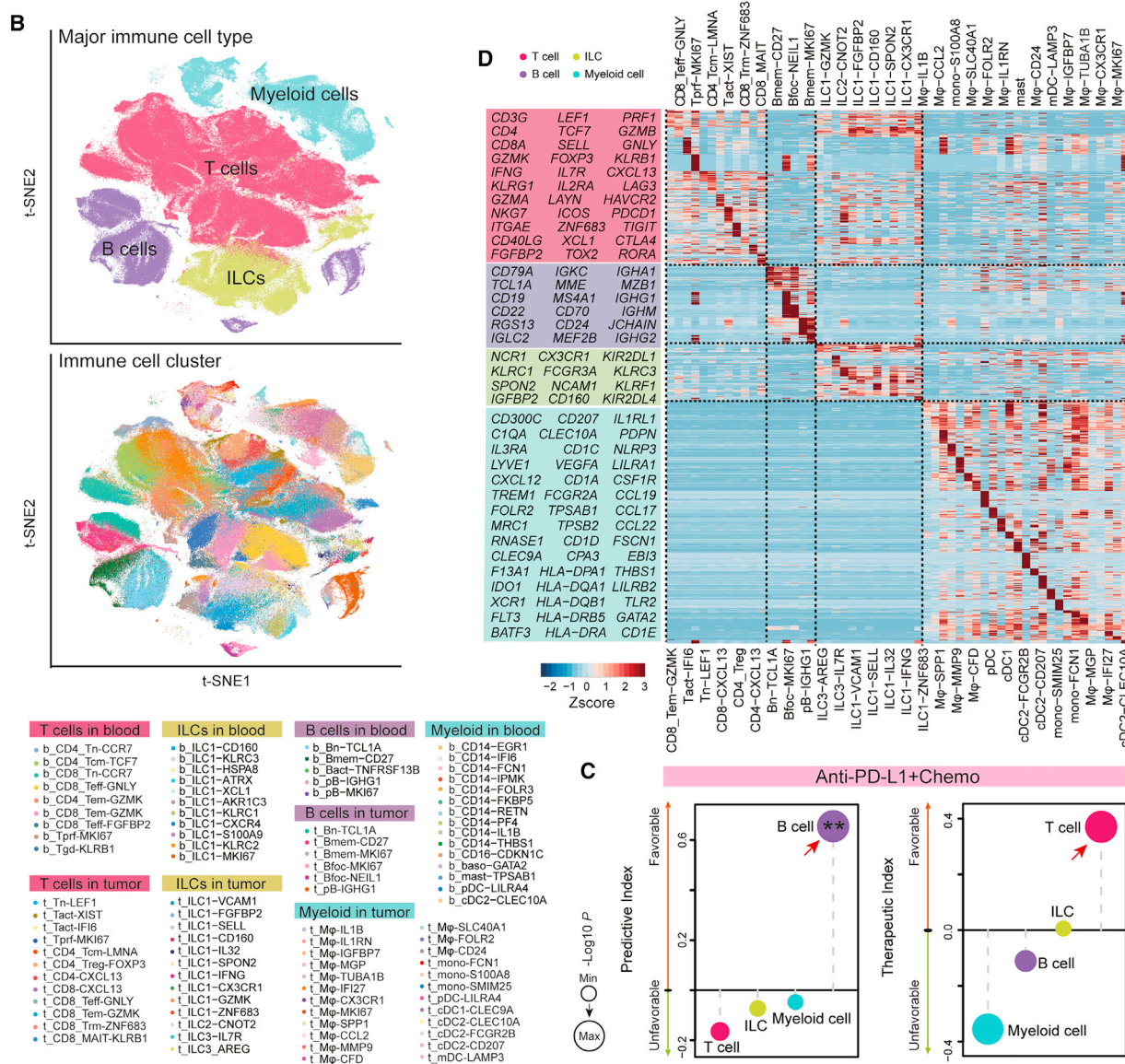
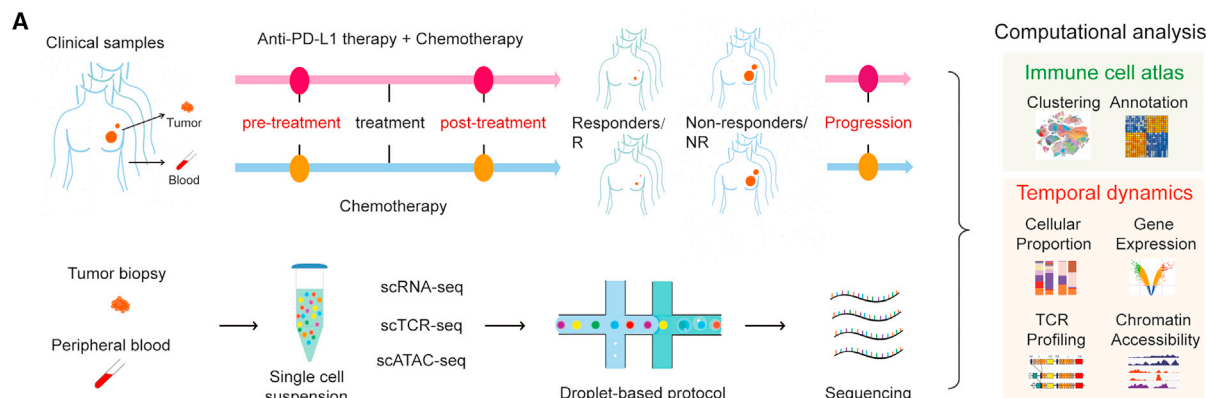
INTRODUCTION

Immune checkpoint blockade (ICB) targeting immunoregulatory pathways such as PD-1/PD-L1 signaling axis has achieved durable responses in a fraction of patients with cancer (Hargadon et al., 2018; Pardoll, 2012; Sharma and Allison, 2015). Despite promising clinical progress, the fundamental mechanisms underlying ICB resistance remain poorly understood. The advances in single-cell technologies enable the comprehensive characterization of the basic properties of tumor-infiltrating immune cells, and a broad spectrum of T cell subpopulations have been identified in the tumor microenvironment (TME) (Azizi et al., 2018; Guo et al., 2018; Li et al., 2019; Savas et al., 2018; Tirosh et al., 2016; Zhang et al., 2018; Zheng et al., 2017a). Notably, T cells in TME often reach a dysfunctional or exhausted state, which is characterized by the overexpression of immune checkpoint molecules and the reduction of effector function (Blank et al., 2019; Wherry, 2011; Zheng et al., 2017a). Although

T cells have been extensively investigated, innate and adaptive immune cells in TME act in concert to control tumor growth (Gajewski et al., 2013), and single-cell studies have highlighted the modulation of other immune cells for T cell responses (Hollern et al., 2019; Zhang et al., 2020). Thus, it is paramount to systematically determine the orchestrated immune responses in anti-tumor immunity and immunotherapies.

Triple-negative breast cancer (TNBC) is an aggressive subtype of breast cancer (BC) with limited therapeutic options (Bianchini et al., 2016). Although ICB shows promise in TNBC treatment, clinical studies have observed a modest response (Dirix et al., 2018; Nanda et al., 2016), which promotes the exploration of ICB in combination with chemotherapy. Indeed, combining nab-paclitaxel with atezolizumab, an anti-PD-L1 antibody, can prolong progression-free survival of patients with metastatic TNBC, as shown by the IMPASSION130 trial and has been approved for the treatment of patients with metastatic or locally advanced unresectable TNBC (Schmid et al., 2018). However,





(legend on next page)

the combination of paclitaxel with atezolizumab fails to meet the primary endpoint in the IMpassion131 trial (Miles et al., 2020), thus presenting an intriguing paradox, although different drug combinations should be evaluated independently. Our incomplete understanding of the underlying mechanisms of such a combination therapy has hindered the improvement of current treatment regimens for metastatic TNBC, underscoring the necessity of deciphering the complex TME and the intertwined effects between combination therapies in patients with TNBC. Single-cell analysis of breast tumor TME following anti-PD-1 treatment has been recently published (Bassez et al., 2021), but it is not clear how such findings can be applied to the anti-PD-L1 studies.

Here, we leverage single-cell RNA-sequencing (RNA-seq) and ATAC-sequencing (ATAC-seq) technologies to characterize the cellular and molecular dynamics of immune cells in 22 patients with TNBC treated with paclitaxel or its combination with atezolizumab. We uncover the underpinnings of resistance and sensitivity to anti-PD-L1 therapies, identify responsive immune cell subtypes to atezolizumab, and suggest the potential drawbacks of paclitaxel in combination with atezolizumab in TNBC treatment.

RESULTS

Immune cell dynamics in patients with advanced TNBC

Twenty-two patients with advanced TNBC were enrolled in this study, half of whom were treated with paclitaxel monotherapy and the other half with paclitaxel plus atezolizumab as first-line therapy. Breast tumors or metastatic tumors were collected according to availability (STAR Methods; Table S1). Tumor-infiltrating lymphocytes (TILs) and PD-L1 have been evaluated for tumors treated with paclitaxel plus atezolizumab, due to their importance in predicting immunotherapy responses. Although responders showed higher levels of TILs and PD-L1, not all tumors with such features responded to anti-PD-L1 therapy (Figures S1A and S1B; Table S1), suggesting the need to dissect the mechanisms of immunotherapies. We thereby resorted to single-cell sequencing to characterize immune features and dynamics in patients with TNBC showing distinct clinical outcomes. In total, 78 tumor biopsies and blood samples were collected at baseline, 4 weeks after treatment initiation, and disease progression (Figure 1A; Table S1). We performed exome sequencing for biopsy samples and found that their genomic alterations were concordant with those commonly identified in human breast tumors from The Cancer Genome Atlas (TCGA) (Koboldt et al., 2012) (Table S1). After quality filtering, we obtained the single-cell transcriptome data for 489,490 high-quality immune cells that included myeloid cells, innate lymphoid cells (ILCs), T cells, and B cells (Figures 1B, S1C, and S1D; Table S2). Interestingly, B cells were mainly enriched in responsive tumors, while myeloid cells were enriched in nonresponsive

tumors of the combination therapy group (Figure S1E). For the dynamic changes of major immune cell types following treatment, B cells decreased while T cells increased after effective treatment in both groups.

We developed two indices, the predictive index (Pi) and the therapeutic index (Ti), to systematically investigate the connection of distinct immune cell types with clinical responses. The Pi measures the correlation between baseline cellular proportions with tumor size changes, while the Ti measures the correlations between cellular proportion dynamics with tumor size changes (STAR Methods). A positive Pi or Ti represents that a higher level of baseline or increase in cellular proportion of the corresponding immune cell subtype is associated with a higher level of decrease in tumor size following treatment, thus predicting or mediating a favorable response, respectively. A negative Pi or Ti represents that the corresponding immune cell subtype predicts or mediates an unfavorable response, respectively.

Based on the Pi analysis, we found that B cells were the most prominent immune cell type that could predict a favorable response to both treatment regimens (Figures 1C and S1F), consistent with recent findings regarding the predictive roles of B cells for ICB response (Cabrita et al., 2020; Helmink et al., 2020; Petitprez et al., 2020). For the Ti analysis, T cells notably exhibited a positive Ti for the combination therapy in contrast to chemotherapy (Figures 1C and S1F), suggesting that atezolizumab may contribute to the effective response through elevating T cells. To generate a comprehensive immune cell atlas in human TNBC, we further reclustered each major immune compartment derived from blood or tumors, separately (STAR Methods). We revealed heterogeneous immune cell subtypes that exhibited distinct molecular signatures, indicative of their unique cellular identities (Figures 1B, 1D, and S1G; Table S2).

CD8-CXCL13 and CD4-CXCL13 expanded in TNBC tumors responsive to the combination therapy

Our aforementioned analyses revealed that T cells might play a central role in PD-L1 blockade therapy. Based on our high-resolution T cell map in TNBC tumors, we identified two T cell clusters (CD8-CXCL13, CD4-CXCL13) that highly expressed *PDCD1* and *CXCL13*; three T cell clusters including naive (Tn), activated (Tact), and proliferative (Tprf) subsets that contained both CD8⁺ and CD4⁺ T cells; and several canonical CD8⁺ or CD4⁺ T cell clusters including regulatory (Treg), central memory (Tcm), effector (Teff), effector memory (Tem), tissue-resident memory (Trm), and mucosal-associated invariant T (MAIT) cell subsets (Figures S2A and S2B; Table S2). These T cell subsets showed no particular tissue preference except for Tn enriched in lymph nodes (LNs) and MAIT enriched in liver metastatic sites (Figure S2C). Of note, CD8-CXCL13 expressed T cell exhaustion-related genes of *TIGIT*, *CTLA4*, and *LAG3*, resembling CD8-LAYN identified by our previous studies (Guo et al., 2018; Zhang et al., 2018; Zheng et al., 2017a), while CD4-CXCL13

Figure 1. Immune cell dynamics in patients with advanced TNBC treated with different regimens

- (A) Schematic overview of the experimental design and analytical workflow.
 (B) t-Distributed stochastic neighbor embedding (t-SNE) visualization of immune cell clusters.
 (C) The Pi and Ti of major immune cell types in TNBC tumors treated with paclitaxel plus atezolizumab. Pi, predictive index; Ti, therapeutic index; Dot size represents the significance evaluated by $-\log_{10}$ (p value). $^{**}0.001 < \text{p value} < 0.01$.
 (D) The expression patterns of signature genes in distinct tumor-infiltrating immune cell clusters. See also Figure S1 and Tables S1 and S2.

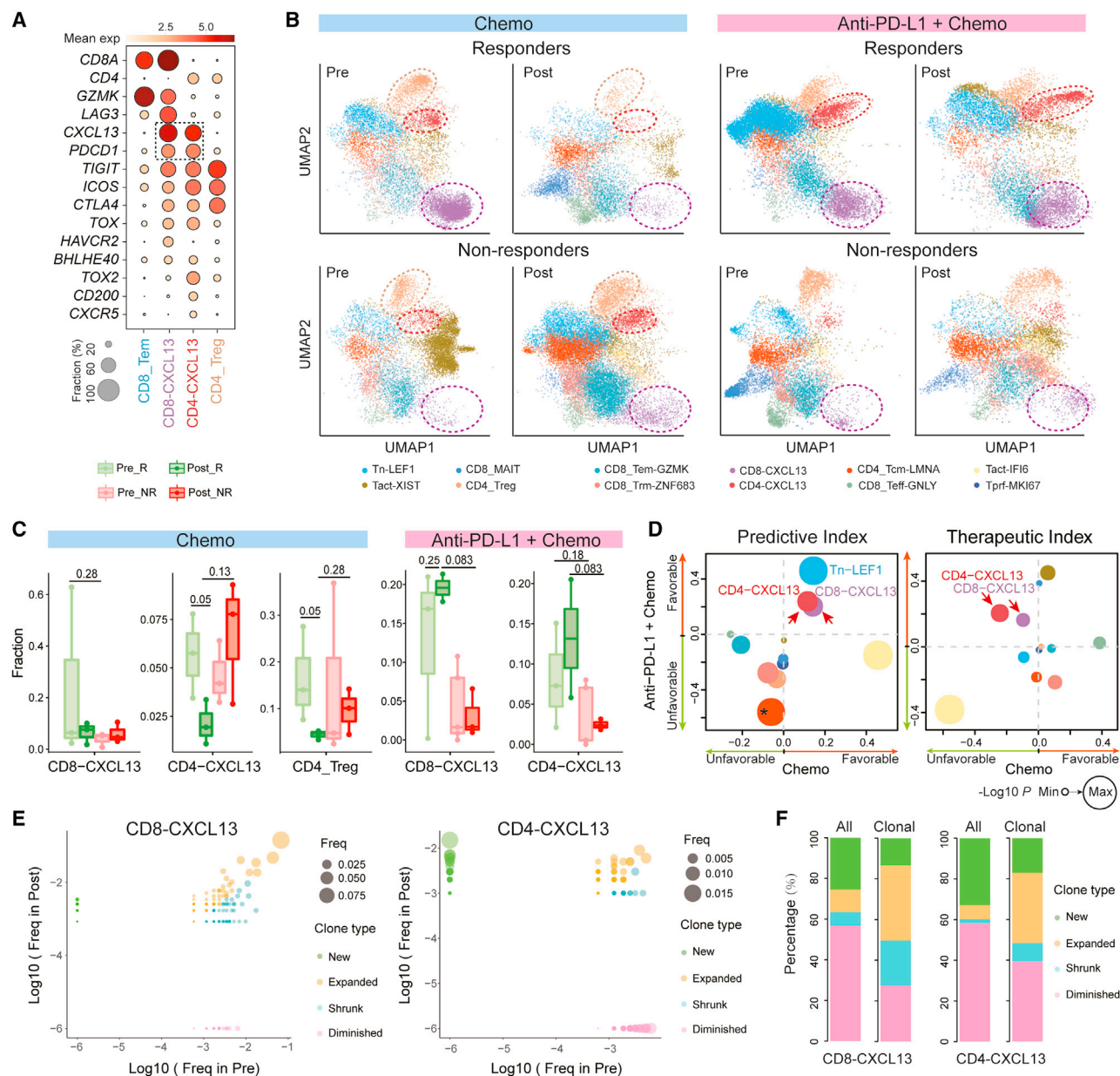


Figure 2. Temporal dynamics of tumor-infiltrating T cell subsets

(A) Expressions of selected genes in tumor-infiltrating T cell clusters.

(B and C) (B) Uniform Manifold Approximation and Projection (UMAP) plots and (C) boxplots showing the temporal alterations of T cell clusters. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 $\times$ the interquartile range. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

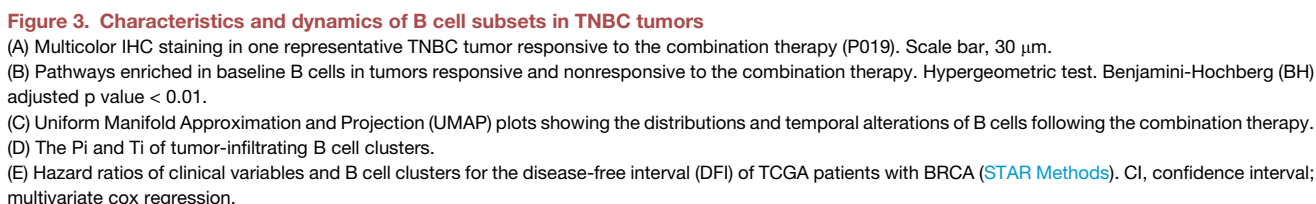
(D) The Pi and Ti of tumor-infiltrating T cell clusters. Dot size represents the geometric mean of $-\log_{10}(p \text{ value})$ in different treatment groups. *left, $0.01 < p \text{ value} < 0.05$ for the combination therapy.

(E) Scatterplots showing TCR clone frequencies of CD8_CXCL13 and CD4_CXCL13 in responsive tumors pre- and posttreatment with the combination therapy. “New” or “Diminished”, TCR clones that emerge or disappear following treatment; “Expanded” or “Shrunk”, TCR clones that exist both pre- and posttreatment, but show higher or lower frequencies following treatment.

(F) Relative percentages of four types of TCR clones in all TCR clones or clonal TCR clones (clone size ≥ 2) for CD8_CXCL13 and CD4_CXCL13 in tumors responsive to the combination therapy. See also Figure S2 and Tables S2 and S3.

not only exhibited features of both T helper type 1 (Th1) cells and T follicular helper (Tfh) cells by expressing *BHLHE40*, *TOX2*, and *CXCR5* (Xu et al., 2019; Zhang et al., 2018), but also expressed exhaustion markers (Figure 2A). The exhaustion features of

CD8_CXCL13 and CD4_CXCL13 implied their tumor-reactivity, since previous studies have shown that tumor-reactive T cells exhibited exhaustion phenotypes due to persistent tumor antigen stimulation (Duhen et al., 2018; Litchfield et al., 2020;



Simoni et al., 2018). Additionally, these two subsets, accompanied with T regulatory cells (Tregs), showed remarkable clonal expansion based on matched T cell receptor (TCR) data, further indicating their antigen-experienced properties (Figure S2A; Table S3).

To decipher the immune cell dynamics following treatment, we first investigated the modulation of paclitaxel on distinct T cell subsets. Compared with nonresponders, responders showed decreased levels of CD8-CXCL13, CD4-CXCL13, and Tregs, accompanied by elevated levels of CD8⁺ Tem, Teff, and CD4⁺ Tcm cells following paclitaxel treatment (Figures 2B, 2C, and S2D). Intriguingly, both CD8-CXCL13 and CD4-CXCL13 expanded in responsive patients following paclitaxel plus atezolizumab treatment, leading to their higher fractions in the combination therapy group compared with paclitaxel alone after treatment (Figures 2B, 2C, and S2E). Similarly, our Ti analysis also demonstrated the expansion of CD8-CXCL13 and CD4-CXCL13 after the combination therapy, in contrast to their reduction after paclitaxel treatment in responsive patients (Figures 2D and S2F). Such observations implied the pertinence of CXCL13⁺ T cells with PD-L1 blockade. We further utilized TCR sequences to trace the lineage origins of CXCL13⁺ T cells in tumors responsive to the combination therapy, and found that their expansion contained both preexisting and newly infiltrated clones (Figures 2E and 2F). In fact, the number of preexisting clones was greater than newly infiltrated clones when considering clonal TCRs (clone size ≥ 2). Thus, considerable preexisting clones contributed to the expansion of CXCL13⁺ T cells following treatment. The association of CD8-CXCL13 with ICB therapies was further confirmed by published data of BC mouse models showing an increase of CD8⁺ PD-1⁺ LAG3⁺ T cells following ICB treatment (Hollern et al., 2019), and also validated by single-cell data of human basal cell carcinoma (BCC) showing an increase of CD8⁺ exhausted T (Tex) cells after ICB treatment (Yost et al., 2019) (Figures S2G and S2H).

CXCL13⁺ T cells and associated B cells predicted favorable responses to anti-PD-L1 immunotherapy

Compared with patients resistant to the combination therapy, responsive patients exhibited higher levels of baseline CD8-CXCL13 and CD4-CXCL13 (Figures 2B and 2C), implying the importance of CXCL13⁺ T cells in shaping clinical outcomes of anti-PD-L1 therapy. Consistently, our Pi analysis also demonstrated that higher levels of CXCL13⁺ T cells could predict better responses to the combination therapy (Figure 2D). In addition to CXCL13⁺ T cells, Tn-LEF1, in spite of its specific enrichment in LNs, was also linked to a favorable response to both treatment regimens (Figure 2D). TCGA-based analysis showed that breast tumors harbored fewer infiltrating CD8⁺ CXCL13⁺ T cells than melanoma and lung tumors (Figure S2I), in line with the lower response rates to ICB for BC (Ribas and Wolchok, 2018; Siu et al., 2017). We further confirmed the presence of CD8-

CXCL13 and CD4-CXCL13 T cells in tumors responsive to the combination therapy by multicolor immunohistochemistry (IHC) staining (Figure 3A).

B cells in TNBC tumors comprised two major subpopulations, CD19⁺ B and plasma B (pB) cells (Figure S3A), and higher levels of baseline CD19⁺ B cells were observed in responders compared with nonresponders for the combination therapy group (Figure S3B). CD19⁺ B cells highly expressed *CXCR5*, the receptor of *CXCL13* (*BCA-1*) (Figure S3A), indicating their potential connections with CXCL13⁺ T cells. Indeed, we observed the colocalization of CD19⁺ B cells with CD8-CXCL13 and CD4-CXCL13 T cells in TNBC tumors responsive to paclitaxel plus atezolizumab (Figure 3A). Such colocalization highly resembled the tertiary lymphoid structure (TLS), as indicated by lymphocyte aggregation, consistent with recent studies suggesting that TLS contributes to immunotherapy responses (Cabrita et al., 2020; Helmink et al., 2020). Additionally, we found that CD19⁺ B cells co-expanded with CD8⁺ Tex cells in responsive patients following ICB treatment in human BCC (Figure S2H) (Yost et al., 2019). Furthermore, B cell makers of *CD79A* and *CD19*, as well as *CXCR5* were highly expressed in TCGA BRCA tumors harboring substantial CD8⁺ CXCL13⁺ T cells compared with those with scarce CD8⁺ CXCL13⁺ T cells (Figure S3C). By comparing transcriptomic differences of B cells between responsive and nonresponsive tumors, we found that B cells enriched in responsive tumors highly expressed genes involved in pathways of antigen processing and presentation, response to interferon (IFN)- γ , and T cell activation. By contrast, B cells enriched in nonresponsive tumors highly expressed genes involved in pathways of B cell activation, immunoglobulin production, and humoral immune response (Figures 3B and S3D). Thus, the function of B cells in antitumor immunity may be mainly attributed to activating CXCL13⁺ T cells.

Bfoc and LTi cells concertedly expanded in TNBC tumors responsive to the combination therapy

We found CD19⁺ B cells comprised heterogeneous subsets of naive B (Bn), memory B (Bmem), Bfoc, proliferating Bmem, and proliferating Bfoc subsets that showed no particular tissue enrichment (Figures S3A, S3E, and S3F; Table S2). Notably, Bn, Bmem, and Bfoc cells were mainly enriched in tumors responsive to paclitaxel plus atezolizumab but not in tumors responsive to paclitaxel (Figures 3C and S3G).

Bfoc cells were characterized by the high expressions of *MEF2B*, *BCL6*, *RGS13*, and *NEIL1* (Figures S3A and S3E) (Basso and Dalla-Favera, 2012; Brescia et al., 2018). Responders harbored substantially higher levels of baseline Bfoc than nonresponders in both treatment groups (Figures 3C and S3G), and our Pi analysis also showed the connection of Bfoc with favorable clinical responses (Figure 3D). For B cell dynamics, we found that Bfoc decreased in responders treated with paclitaxel, but increased in those treated with paclitaxel plus atezolizumab

(F) Correlations of distinct immune cell subsets with Bfoc in their cellular proportions in posttreatment patients with TNBC. PCC, Pearson correlation coefficient.

(G) Heatmap showing the expression of ligand-receptor pairs highly expressed in Bfoc, CD4-CXCL13, or CD8-CXCL13.

(H) Correlations of distinct immune cell subsets with Bfoc in their cellular proportion dynamics in TNBC patients.

(I) Boxplots showing the temporal alterations of LTi-like ILC3 cluster. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 $\times$ the interquartile range. Kruskal-Wallis test.

(J) The Pi and Ti of tumor-infiltrating ILC clusters. Pi, predictive index; Ti, therapeutic index. See also Figure S3 and Table S2.

(Figures 3C, 3D, and S3G–S3I), suggesting the pertinence of Bfoc cells to atezolizumab treatment. Importantly, we characterized the prognostic significance of distinct B cell subsets, and revealed that Bfoc was the most significant B cell subtype associated with favorable clinical outcomes for TCGA patients with BRCA (Figure 3E; STAR Methods). Therefore, Bfoc appeared to be the most important B cell subset for patient survival and ICB-immunotherapy.

We next interrogated the connection of Bfoc with other immune cells. By delineating the correlated immune cell subtypes, we found that CD4-CXCL13 and CD8-CXCL13 exhibited high correlations in cellular proportions with Bfoc, especially in the posttreatment group (Figures 3F and S3J). Such connections indicated that Bfoc may interact with CXCL13⁺ T cells to play a role in antitumor immunity. Focusing on ligand-receptor pairs, we found that *CXCR5* showed the highest expression level in the Bfoc subset, and multiple pairs including CD40LG-CD40, interleukin (IL)21-IL21R, and IL6ST-EBI3 might mediate the crosstalk between Bfoc and CD4-CXCL13 (Figure 3G). Interestingly, LT α -like ILC3 (ILC3-IL7R), featured by the expressions of *CCR7*, *IL7R*, *GPR183*, *LTB* (lymphotoxin), and *IL23R*, also exhibited high correlation with Bfoc and concertedly expanded with Bfoc after the combination therapy but not after chemotherapy (Figures 3H–3J, S3J, and S3K). Of note, ILC3-IL7R could predict a favorable response to both treatment regimens (Figure 3J), indicating its potential role in effective antitumor immunity.

cDC1 expanded in TNBC tumors responsive to the combination therapy

The importance of CXCL13⁺ T cells in anti-PD-L1 therapy prompted us to investigate how they might be activated and enriched in certain patients with TNBC. Since myeloid cells in TME can potentially modulate T cell responses (Lavin et al., 2017; Zhang et al., 2020), we next focused on the myeloid compartment in TNBC tumors. While our data did not cover granulocytes possibly due to technical biases, we observed DCs, macrophages, monocytes, and mast cells. Among these major subpopulations, we observed higher levels of DCs in responders compared with nonresponders both at baseline and after treatment with the combination therapy but not chemotherapy (Figures 4A and S4A), supporting their roles in ICB therapy. To determine the specific subsets of DCs in TNBC tumors, we performed unsupervised clustering and identified plasmacytoid DC (pDC), cDC1, mature DC (mDC, or LAMP3⁺ DC) featured by *CCR7*, *LAMP3*, *CCL19*, *IDO1*, and *PD-L1*, and three subsets of conventional DC2 (cDC2) that included Langerhans cells (LCs, cDC2-CD207), proinflammatory and anti-inflammatory cDC2 (cDC2-CLEC10A, cDC2-FCGR2B), based on their signature genes (Figures S4B–S4D; Table S2).

Interestingly, higher levels of baseline cDC1 and pDC could suggest favorable responses to paclitaxel, but none of these DC subsets could indicate favorable responses to the combination therapy, likely hinting that DC subsets were not key determinants of effective anti-PD-L1 therapy (Figure S4E). For distinct DC subsets, we next examined their temporal dynamics in different treatment groups. Compared with nonresponders, responders showed decreased levels of mDC after paclitaxel therapy, a temporal pattern resembling those of potential tu-

mor-reactive T cells in the same treatment group (Figures 2C and 4B). Although cDC1 and proinflammatory cDC2 could both transit into mDC based on RNA velocity analysis (Figure S4F), consistent with findings in human liver and lung cancer (Cheng et al., 2021; Maier et al., 2020; Zhang et al., 2019), mDC was tightly connected with cDC1 in TNBC tumors, due to their transcriptional similarity and correlated dynamic changes in cellular proportions (Figures 4C and 4D). For patients treated with paclitaxel plus atezolizumab, we also observed that mDC decreased in responders following treatment in contrast to the increase of cDC1 and pDC, likely due to the selective reduction of mDC by paclitaxel regimen (Figure 4B). Importantly, higher levels of cDC1, mDC, and pDC were found in responsive patients after treatment with paclitaxel plus atezolizumab compared with paclitaxel alone, indicating their involvement in anti-PD-L1 therapy (Figure S4G). cDC1 and pDC were further shown to be associated with the favorable response to the combination therapy but not chemotherapy by our Ti analysis (Figure 4E). Notably, the reduction of mDC following paclitaxel treatment was subdued when atezolizumab was also used, indicating its pertinence to PD-L1 blockade.

Of note, both mDC and cDC1 exhibited coordinated temporal dynamics with CD4-CXCL13 (Figures 4D and S4H), and we confirmed their correlations by the bulk RNA-seq data of TCGA BRCA (Figure 4F). In addition, we revealed multiple ligand-receptor pairs including CD40LG-CD40 that might mediate the crosstalk among mDC, cDC1, and CD4-CXCL13 (Figure 4G). Interestingly, CD8-CXCL13 also exhibited high correlations with mDC, cDC1, and CD4-CXCL13 based on their temporal dynamics in cellular proportions (Figures 4D and S4H). Such observations implied that CD4-CXCL13 and CD8-CXCL13 might be both activated by cDC1 or mDC, as hinted by recent studies showing that cognate CD4⁺ T cells license cDC1 to induce CD8⁺ T cell immunity (Cancel et al., 2019; Ferris et al., 2020).

The phenotypes of macrophages were associated with T cell states

Focusing on the monocyte/macrophage compartment, we found that nonresponders exhibited higher baseline levels of macrophages (M ϕ) compared with responders in the combination therapy group, and macrophages enriched in nonresponders highly expressed *SPP1*, *TREM2*, *FN1*, and *C3*, while those enriched in responders highly expressed *CXCL9*, *MMP9*, and *PLA2G2D* (Figures 4A and 5A; Table S4). Notably, *TREM2* and *SPP1* have been shown to be expressed in anti-inflammatory tumor-associated macrophages (TAMs) (Molgora et al., 2020; Zhang et al., 2020). Further unsupervised clustering revealed extensive heterogeneity, giving rise to multiple subsets of monocytes and macrophages (Figures S5A–S5D). Monocyte subsets highly expressed *FCN1*, *S100A8*, and *S100A9*, while macrophage subsets highly expressed *CD68*, *APOE*, and *C1Qs*, resembling features of TAMs (Lavin et al., 2017; Zhang et al., 2019). Notably, macrophage subsets in TNBC tumors tended to coexpress canonical M1 and M2 signatures (Figure S5E), consistent with recent findings in human breast and lung cancer (Azizi et al., 2018; Lavin et al., 2017).

To understand how CXCL13⁺ T cells were modulated by other immune cells, we utilized the TCGA BRCA data to examine the transcriptional features of breast tumors showing high

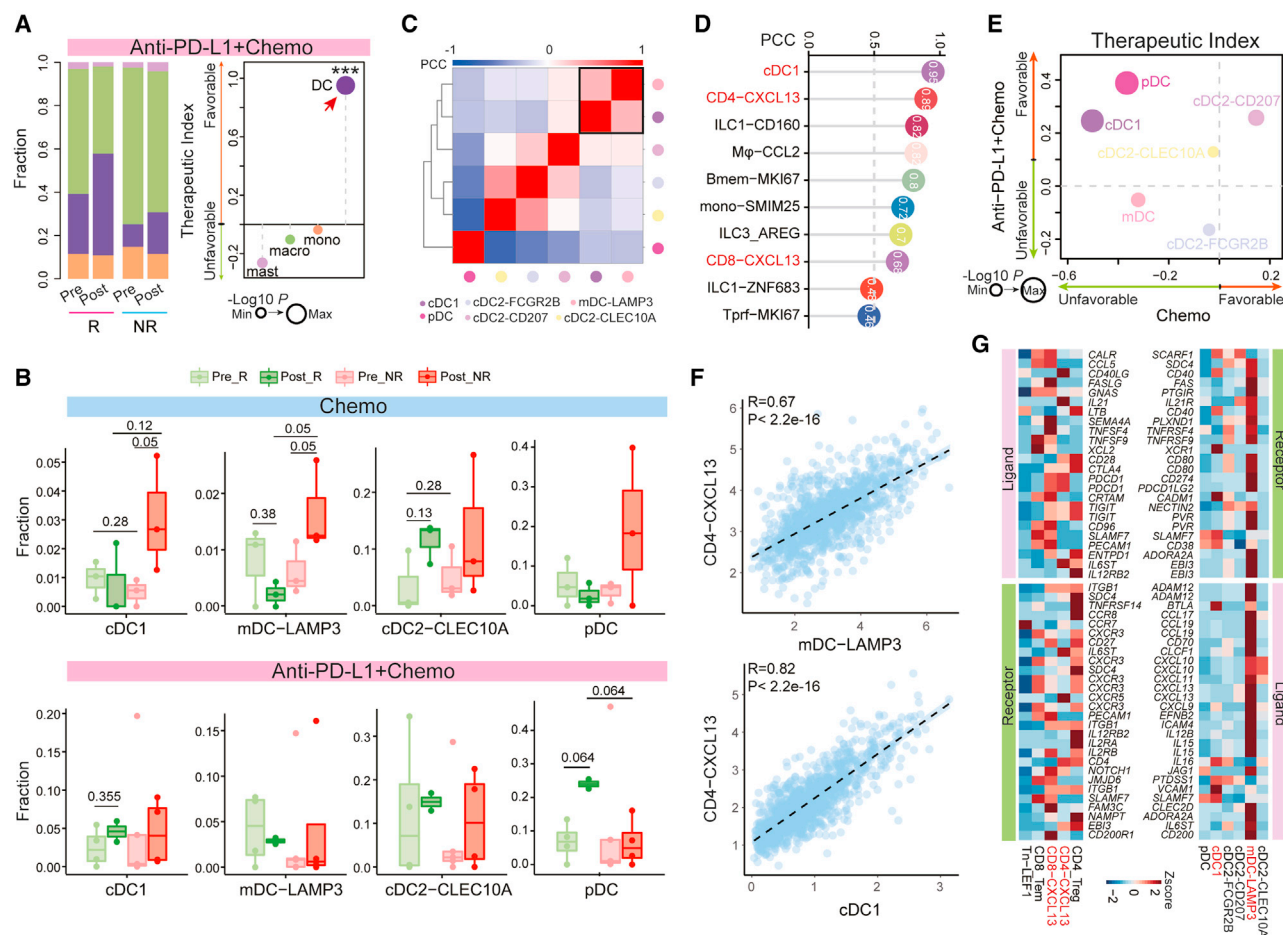


Figure 4. Temporal dynamics of DCs in TNBC tumors

(A) The cellular compositions and Ti of major myeloid subpopulations in TNBC tumors treated with the combination therapy. (B) Boxplots showing the temporal alterations of DC clusters in TNBC tumors. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 × the interquartile range. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test. (C) Correlations among different DC subsets based on their marker gene expressions. PCC, Pearson correlation coefficient. (D) Correlations of distinct immune cell subsets with mDC in their cellular proportion dynamics in TNBC tumors treated with paclitaxel. (E) The Ti of DC clusters in TNBC tumors. (F) Scatterplots showing correlations of CD4-CXCL13 with mDC and cDC1 in TCGA BRCA data based on their signature gene expressions (STAR Methods). Pearson correlation test. (G) Heatmap showing the expression of ligand-receptor pairs highly expressed in mDC, cDC1, CD4-CXCL13, or CD8-CXCL13. See also Figure S4 and Table S2.

expressions of *CD8A* and *CXCL13* (Figure S2I). Expectedly, a group of immune-related genes were identified (Figure 5B; Table S4). To estimate the presence of concomitant immune cell subtypes with potential tumor-reactive T cells, we examined the expressions of the above genes, after excluding T cell and B cell markers, in our myeloid cell subsets. Notably, two macrophage subsets, Mφ-CCL2 and Mφ-MMP9, together with mDC and other DC subtypes, were among the top-ranked immune subtypes (Figure 5C), suggesting their connections with CXCL13⁺ T cells. The Mφ-MMP9 subset highly expressed *PLA2G2D*, *IL2RG*, and *CXCL9*, while Mφ-CCL2 highly expressed *CXCL9*, *CXCL10*, and *CXCL11* (Figure 5D). CXCL9 and CXCL10 are chemokines known to mediate T cell recruitment (House et al., 2020; Peperzak et al., 2013), and their high expressions in Mφ-MMP9 and Mφ-CCL2 supported the proinflammatory properties of these two macrophage subsets.

In addition to the above transcriptome-based analyses, we also examined immune cell proportions, and found that Mφ-CCL2 and Mφ-MMP9 were also highly correlated with CXCL13⁺ T cells (Figure 5E). Of note, higher baseline levels of these two macrophage subsets suggested a favorable response to the combination therapy (Figures 5F, S6A, and S6B). NicheNet analyses revealed that the key signature genes of Mφ-CCL2 and Mφ-MMP9, including *CXCL9* and *CXCL10*, were associated with *IFNG* and *TNF* (Figure 5G; Table S4), indicating that the proinflammatory phenotypes of macrophages might be sculpted by active antitumor immune responses in TME. Consistently, compared with nonresponders, we observed the simultaneous enrichments of Mφ-MMP9 and CD8-CXCL13 in tumors responsive to paclitaxel plus atezolizumab (Figures 5H and S6C).

In contrast to Mφ-CCL2 and Mφ-MMP9, the baseline levels of Mφ-MKI67, Mφ-SLC40A1, and Mφ-MGP were positively

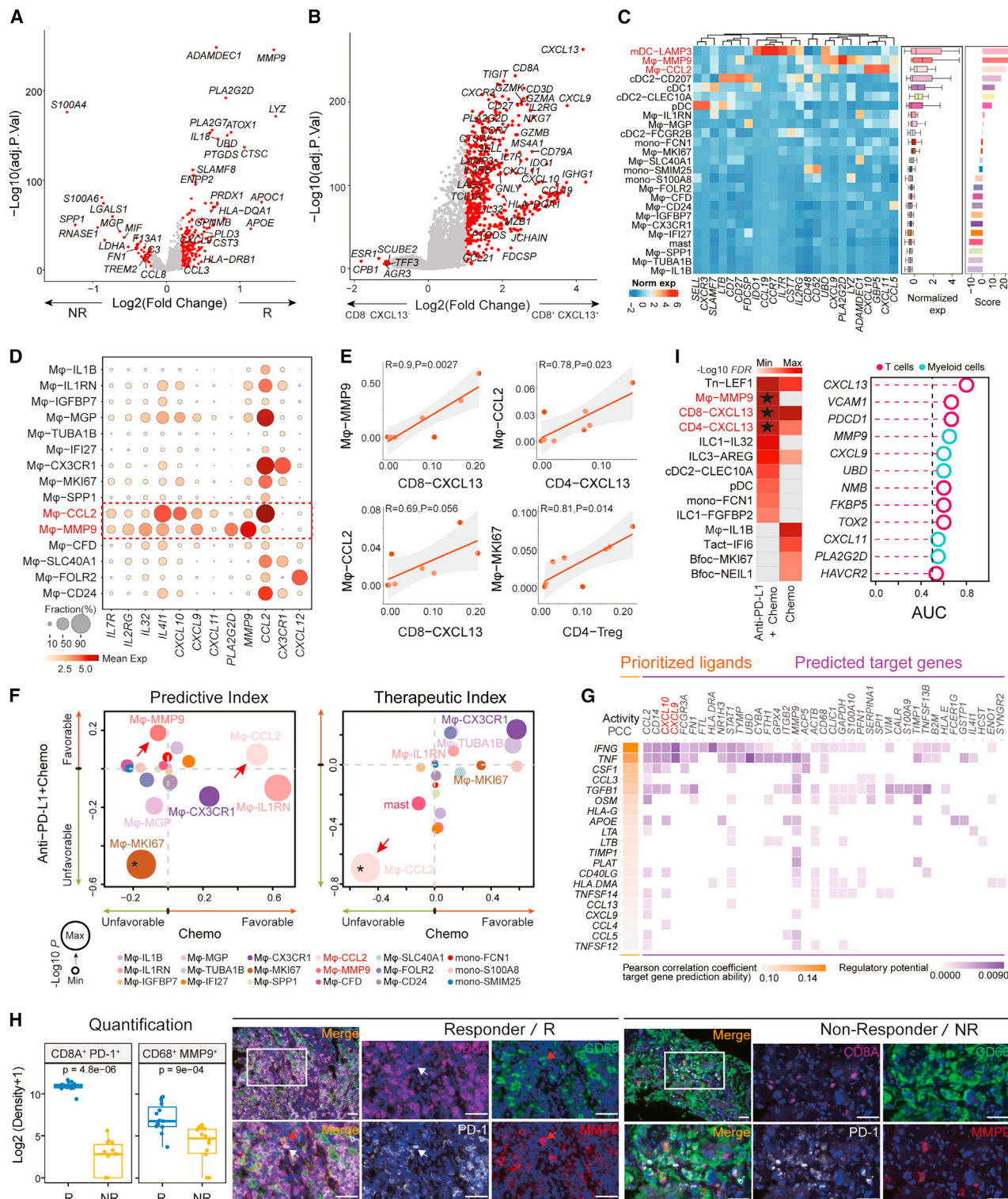


Figure 5. The connections of macrophage subsets with CXCL13⁺ T cells in TNBC tumors

(A and B) (A) Volcano plots showing differentially expressed genes (DEGs) of macrophages in pretreatment responsive and nonresponsive tumors of the combination therapy group, and (B) DEGs of TCGA patients with BRCA grouped by the expression levels of CD8A and CXCL13 in Figure S2I. BH adjusted p value < 0.01. Limma moderated t test.

(C) The expression levels of selected genes highly expressed in TCGA patients with BRCA showing high levels of CD8A and CXCL13 (Figure S2I) in our single-cell myeloid subsets. Score means the sum of normalized expressions.

(legend continued on next page)

correlated with Tregs, whereas M ϕ -CX3CR1, M ϕ -TUBA1B, and M ϕ -FOLR2 were negatively correlated with CXCL13⁺ T cells (Figure S6A). These subsets highly expressed *SPP1* and *TREM2* (Figure S5D), resembling SPP1⁺ TAMs identified by our previous study in human colorectal cancer (Zhang et al., 2020), and thus represented anti-inflammatory macrophages. We observed their enrichments in nonresponsive tumors to the combination therapy based on our Pi analysis (Figure 5F). For the dynamic changes of macrophage subsets following different treatments, we found that the paclitaxel regimen selectively decreased proinflammatory macrophages of M ϕ -CCL2 and M ϕ -MMP9, but increased immunosuppressive macrophages in responsive patients, while atezolizumab seemed to have little effect on macrophage subsets (Figures 5F and S6B). These findings indicated that paclitaxel may confer an immunosuppressive TME.

To facilitate the development of applicable biomarkers in clinical practice, we further evaluated the predictive performance of key signature genes of CXCL13⁺ T cells and M ϕ -MMP9 by constructing scoring classifiers, and revealed a set of markers showing high predictive potentials (Figure 5I; STAR Methods). Thus, the key signature genes of core antitumor immune subsets could be explored as biomarkers to predict ICB responses.

Interestingly, we found that higher levels of myeloid cells in blood, in which monocytes were largely predominant, could predict a favorable response to both treatment regimens (Figures S6D and S6E). Further dissecting the total transcriptomic features of baseline blood-derived myeloid compartment revealed that suppressive TAM-associated genes, including *CD163*, *VCAN*, and *THBS1*, were enriched in nonresponders while *CD74* and HLA class II genes were enriched in responders of the combination therapy group (Figure S6F; Table S4). So it appeared that monocytes in blood could partially reflect the immune features in TME. Of note, we did not detect cDC1 in blood, but instead observed the increase of cDC2 following the combination therapy in responsive tumors (Figure S6G).

CD8-CXCL13 showed enhanced cytotoxicity following anti-PD-L1 therapy

Considering the importance of CXCL13⁺ T cells in anti-PD-L1 therapy, we next examined their treatment-induced transcriptional dynamics. Notably, CD8-CXCL13 showed lineage connection with CD8⁺ Tem cells based on RNA velocity and TCR sharing analyses (Figures S7A–S7C). Transcriptome analyses indicated that effector and memory-related genes of *IFNG*, *GZMK*, *GZMA*, and *CD44*; transcription factors (TFs) of *TBX21*, *BHLHE40*, and *BCL6*; co-stimulatory molecules of *CD28* and *ICOS*; as well as HLA genes and integrins were upre-

gulated while exhaustion-related genes, including *HAVCR2*, *LAYN*, *TIGIT*, and *LAG3*, were downregulated in CD8-CXCL13 after treatment (Figures 6A and S7D; Table S5). The phenotypic shift of CD8-CXCL13 was also reflected by its decreased exhaustion score and increased effector memory score (Figure 6B). In addition, genes upregulated in CD8-CXCL13 after treatment were involved in pathways of positive regulation of T cell-mediated cytotoxicity, antigen processing and presentation, and IFN- γ -mediated signaling pathway (Figure S7E), further suggesting its augmented effector property after PD-L1 blockade treatment.

We additionally performed single-cell ATAC-seq (scATAC-seq) assays to examine changes in chromatin accessibility, and obtained such data for 30,758 immune cells from five samples, including paired samples derived from responsive patients treated with paclitaxel plus atezolizumab (Table S6). Immune cell subtypes were identified based on our integrative analyses of scATAC-seq and single-cell RNA-seq (scRNA-seq) data (STAR Methods). The chromatin accessibility of signature genes for different subsets confirmed their cellular identities (Figure S7F; Table S7). The epigenetic features of CD8-CXCL13 cells, which showed specific enrichments of peaks in T cell exhaustion markers, including *PDCD1*, *LAG3*, and *CTLA4*, were consistent with those of CD8⁺ Tem cells characterized by previous studies in mouse models, human melanoma, and non-small-cell lung cancer (Philip et al., 2017; Sen et al., 2016) (Figures 6C and S7G). Compared with other T cell subtypes in their chromatin accessibility profiles, CD8-CXCL13 and CD8⁺ Tem were more similar to each other, which confirmed their connections in a continuous spectrum of lineage differentiation (Guo et al., 2018; Zhang et al., 2018; Zheng et al., 2017a). Specifically, both CD8-CXCL13 and CD8⁺ Tem harbored open chromatin regions (OCRs) in the *IFNG*, *GZMK*, and *PDCD1* loci. Nevertheless, CD8-CXCL13 exhibited a distinct cell state from CD8⁺ Tem, due to their enriched OCRs in *CXCL13*, *CTLA4* and *LAYN* (Figures 6C and S7G).

We next examined the global dynamics of chromatin accessibility for CD8⁺ Tem, CD8-CXCL13, and CD4-CXCL13, by comparing the paired scATAC-seq data before and after the combination therapy. Characterization of the differential OCRs and related genes in each T cell subset informed the augmented effector and memory properties of CD8-CXCL13 cells, which showed enhanced accessibilities in *CD44*, *CXCR4*, *KLRG1*, *XCL2*, *CST7*, *ICOS*, and *CD28*, while diminished accessibilities in exhaustion-related genes, such as *CREM* and *MYO7A*, following treatment (Figure 6D; Table S7). Additionally, the overrepresented TF binding motifs (TFBMs) in differentially accessible peaks also confirmed the phenotypic change of CD8-CXCL13, with TFBMs of *CREM* and *RBPJ* tending to be

(D) The expression patterns of selected genes in tumor-infiltrating macrophage subsets.

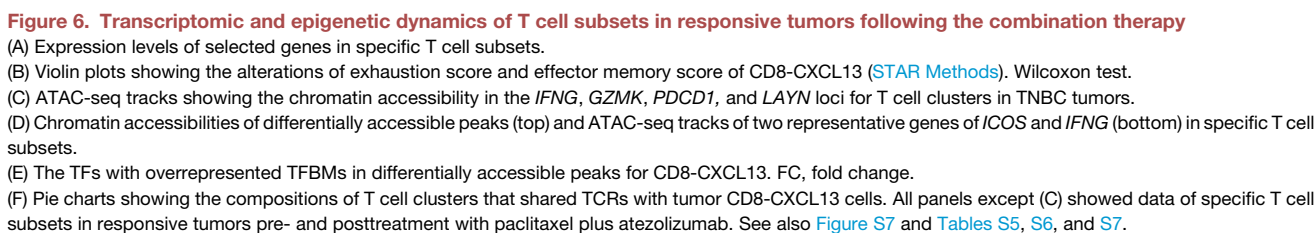
(E) Scatterplots showing correlations of selected macrophage subsets with T cell subsets in TNBC tumors treated with the combination therapy. Pearson correlation test.

(F) The Pi and Ti of tumor-infiltrating macrophage/monocyte subsets.

(G) Ligand activities and predicted target genes identified by NicheNet analysis based on the key signature genes of M ϕ -MMP9 and M ϕ -CCL2 subsets.

(H) Multicolor IHC staining in one representative responder (P012) and one nonresponder (P017) treated with the combination therapy. Scale bar, 30 μ m. Each dot in boxplots represents one view. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 $\times$ the interquartile range. Wilcoxon test.

(I) The enrichments of immune cell subsets in responsive tumors, and the predictive performance of key signature genes of CXCL13⁺ T cells and M ϕ -MMP9 for the combination therapy (STAR Methods). Chi-square test; FDR, false discovery rate. Genes with an area under curve (AUC) $\geq$ 0.5 are shown. See also Figures S5 and S6 and Tables S2 and S4.



1588 Cancer Cell 39, 1578–1593, December 13, 2021

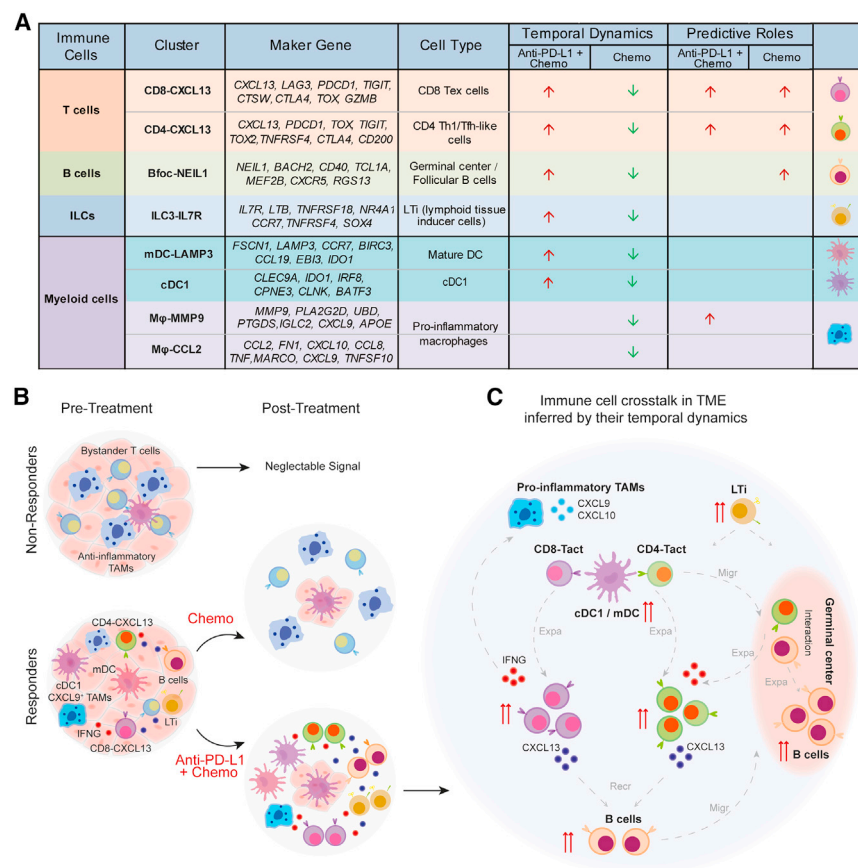


Figure 7. Summary of immune cell features and dynamics in TNBC tumors

(A) Characteristics and dynamics of key immune cell subsets following different treatments. Red (or green) arrows represent higher baseline levels predicting favorable (or unfavorable) responses, or the increase (or decrease) of immune cell subsets in responsive patients following treatment.

(B) Immune features in responsive and non-responsive tumors and their dynamics following different treatment regimens. TNBC tumors with substantial baseline CXCL13⁺ T cells, B cells, and proinflammatory macrophages would show sensitivity, in contrast to resistance of TNBC tumors with substantial immunosuppressive macrophages, to the combination therapy. Paclitaxel regimen decreased while atezolizumab increased key antitumor immune cells in responsive tumors. (C) Potential crosstalk between CXCL13⁺ T cells and their correlated immune cells inferred based on treatment-induced temporal dynamics. Expa, expansion; Migr, migration; Recr, recruitment.

baseline, but decreased following the combination therapy, possibly due to the impact of paclitaxel (Figure S7H). Of note, we did not observe CXCL13⁺ T cells in blood, but found that blood CD8⁺ Tem cells shared higher levels of TCR clones with tumor CD8-CXCL13, and also exhibited elevated migration index (Zhang et al., 2018) in responsive patients following the combination therapy (Figures 6F, S7I, and S7J). Such observations indicated that detectable systemic immune response was induced in responsive patients following PD-L1 blockade treatment.

DISCUSSION

Here, we investigated the temporal dynamics of immune cells in patients with advanced TNBC treated with paclitaxel or its combination with atezolizumab, and uncovered the immune features associated with patient responses (Figures 7A–7C). Our analyses indicated that Bfoc but not pB cells expanded concertedly with LTi cells, CXCL13⁺ T cells, and cDC1 following paclitaxel plus atezolizumab treatment. Notably, cDC1 has been recently proven to function as antigen-presenting cells to prime both CD4⁺ and CD8⁺ T cells and directly orchestrate their crosstalk for achieving optimal antitumor immunity (Ferris et al., 2020). Thus, B cells might contribute to antitumor immunity by interacting with and activating CD4-CXCL13 that could help prime CD8-CXCL13 through cDC1. Although TLS has been shown to exhibit predictive roles, their importance

should be considered in conjunction with the TILs. Our analyses informed the cellular crosstalk among key immune cell subsets in TLS and suggested the underlying mechanisms of the predictive importance of TLS for immunotherapies. Further, our analyses revealed that high levels of CXCL9 and CXCL10, two chemokines recently shown to be linked to effective ICB responses (House et al., 2020; Litchfield et al., 2020; Qu et al., 2020), could indicate the presence of CXCL13⁺ T cells in TME.

Our analyses revealed that the paclitaxel regimen could selectively reduce key antitumor immune cells but elevate immunosuppressive macrophages in TNBC tumors, and thus might not benefit the accompanying atezolizumab. Although the immune cell dynamics could also be affected by disease progression during treatment, the consistent patterns in responders in contrast to non-responders suggested their relevance to the paclitaxel regimen. The reduction of key antitumor immune cells might be caused by the direct and selective damage by paclitaxel, likely due to the proliferative states of potential tumor-reactive T cells (Li et al., 2019; Zhang et al., 2018). Of note, steroid premedication is a necessary procedure for the paclitaxel regimen, and it has been reported that steroid premedication could weaken antitumor immune responses (Arbour et al., 2018; Hegde and Chen, 2020; Oles et al., 2016). Further investigations are needed to evaluate the impact of paclitaxel on immune cells after excluding the potential immune suppression by steroid premedication. In addition, different chemotherapeutic agents might lead to distinct properties of TME that may influence responses to accompanying checkpoint inhibitors. Further investigations are needed to illustrate the mechanistic distinctions of paclitaxel and nab-paclitaxel when combined with atezolizumab for TNBC treatment.

The two different treatment regimens in our study could induce opposite dynamic changes of key antitumor immune subsets only in responsive patients who had high levels of baseline CXCL13⁺

T cells. Thus, the intrinsic properties of TNBC tumors, especially the baseline level of CD8-CXCL13 T cells, seem to be the key determinants of clinical responses and the dynamic patterns of immune cells following different treatments. We reason that for patients with high levels of baseline CXCL13⁺ T cells, they can tolerate the deleterious effect of paclitaxel regimen, and thus become responders and would further increase the levels of CXCL13⁺ T cells after the combination therapy. For patients with low levels of CXCL13⁺ T cells, the effect of atezolizumab would fail to overcome the deleterious effect of paclitaxel regimen. Of note, although our findings suggested that paclitaxel may compromise the efficacy of accompanying atezolizumab, it is unclear whether and how paclitaxel will affect other ICB antibodies, since distinct combinations may lead to substantial differences, and each combination should be considered in the right context.

Limitations of our study include the small sample size and distinct biopsy sites. The small sample size limited the statistical significance of our results. In addition, breast tumors were collected in the chemotherapy group, while metastatic tumors were collected in the combination therapy group since these patients had already gone through breast tumor surgical removal. Immune infiltration levels are in general different in distinct metastatic sites, which may affect the clinical outcomes of immunotherapies. Further studies are needed to investigate the immune features in patients after controlling such variables. Of note, the immune cell dynamics in our study are evaluated based on their relative changes in site-matched samples, thus theoretically counteracting the influence of distinct tissue origins. Additionally, limitations of our study are partially circumvented by introducing the Pi and the Ti to investigate concordant patterns between immune cell compositions and clinical outcomes in all patients. These two indices utilized the relative changes of tumor sizes for all patients in a continuous axis and thus avoided the gross classification of responders and nonresponders. Although immune cells might exhibit tissue preference (Cheng et al., 2021; Zhang et al., 2018), major conclusions in our study were made based on immune cells showing no particular tissue enrichment, and main findings have been confirmed by public data and IHC experiments.

Despite these limitations, our study invokes additional key questions. For instance, patients with LN metastases showing high levels of Tn cells were responsive to the combination therapy, suggesting the potential roles of LN metastasis and Tn cells in predicting favorable responses to ICB treatment. Interestingly, monocytes in the blood of these responsive patients showed a proinflammatory phenotype, and proliferative T cells were specifically enriched in the blood of these patients compared with those nonresponsive patients, hinting systemic immune activation. Our findings are consistent with previous studies (Krieg et al., 2018; Spitzer et al., 2017), and future studies are needed to show how LN metastasis is associated with the activation of systemic immunity following ICB therapies. Additionally, although we have seen promising trends for the link between immune cells in blood and those in tumors, further comprehensive studies will be needed to establish the connections between blood and solid tumors, which may confer the development of blood-derived immune biomarkers.

In conclusion, our study illuminates key immune cell subsets during ICB treatment, and provides new insights into the cellular underpinnings of distinct responses to PD-L1 blockade in pa-

tients with metastatic TNBC, although further investigations are needed to validate our findings. The compendium of transcriptomic and epigenetic data, coupled with TCR sequences, will provide a rich resource for understanding the functional states and dynamics of immune cells following different treatment regimens in advanced TNBC. An interactive portal (http://TNBC_PD-L1.cancer-pku.cn) has been developed to facilitate the usage of our data for the wide research community.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **RESOURCE AVAILABILITY**
 - Lead contact
 - Materials availability
 - Data and code availability
- **EXPERIMENTAL MODEL AND SUBJECT DETAILS**
 - Human specimens
- **METHOD DETAILS**
 - Sample collection and processing
 - Single-cell RNA-seq library and TCR-seq library preparation and sequencing
 - Single-cell ATAC-seq library preparation and sequencing
 - Bulk DNA and RNA isolation and sequencing
 - SP142 PD-L1 immunohistochemistry
 - TILs evaluation
 - Multi-color immunohistochemistry
 - Single-cell RNA-Seq data processing
 - Dimension reduction and unsupervised clustering for scRNA-seq data
 - Single-cell ATAC data processing
 - Dimension reduction and unsupervised clustering for scATAC-seq data
 - TCR sequences assembly
 - Bulk RNA-sequencing and exome-sequencing data processing
 - Definition of predicative index and therapeutic index
 - Tissue enrichment and time-point distribution analysis
 - Definition of exhaustion score and effector memory score
 - Survival analysis
 - Correlations analysis
 - Predictive performance evaluation for signature genes
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
 - Statistical analysis
- **ADDITIONAL RESOURCES**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ccell.2021.09.010>.

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AUTHOR CONTRIBUTIONS

Y.Y.Z., H.C., H.M., and X.H. conceptualized the study; Y.Y.Z., H.C., H.M., X.H., R.G., and B.L. performed the methodology; Y.Y.Z., H.C., H.M., and X.H. performed the study investigation; Y.Y.Z., and Z.Z. wrote the original draft; Y.Y.Z., Z.Z., Z.L., F.M., B.X. H.C., and H.M. performed the review and editing; Z.L., Z.Z., F.M., B.X., H.C., H.M., and Y.Y.Z. acquired the funding; Y.H.Z., L.N., X.S., Y.W., Q.C., X.Y., T.G., T.H., X.G., and T.Q. obtained resources; Z.Z., Z.L., F.M., and B.X. supervised the study.

DECLARATION OF INTERESTS

Z.Z. is a founder of Analytical Biosciences and a member of its scientific advisory board. Z.Z. is also a consultant for GennLife and InnoCare Pharma. X.H. and R.G. are employees of Analytical Biosciences. All financial interests are unrelated to this study. The other authors declare no competing interests.

INCLUSION AND DIVERSITY

We worked to ensure ethnic or other types of diversity in the recruitment of human subjects. One or more of the authors of this paper received support from a program designed to increase minority representation in science. While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list.

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REFERENCES

Arbour, K.C., Mezquita, L., Long, N., Rizvi, H., Auclin, E., Ni, A., Martinez-Bernal, G., Ferrara, R., Lai, W.V., Hendriks, L.E.L., et al. (2018). Impact of baseline steroids on efficacy of programmed cell death-1 and programmed death-ligand 1 blockade in patients with non-small-cell lung cancer. *J. Clin. Oncol.* **36**, 2872–2878.

Azizi, E., Carr, A.J., Plitas, G., Cornish, A.E., Konopacki, C., Prabhakaran, S., Nainys, J., Wu, K., Kisieliovas, V., Setty, M., et al. (2018). Single-cell map of diverse immune phenotypes in the breast tumor microenvironment. *Cell* **174**, 1293–1308.e1236.

Bassez, A., Vos, H., Van Dyck, L., Floris, G., Arijis, I., Desmedt, C., Boeckx, B., Vanden Bempt, M., Nevelsteen, I., Lambein, K., et al. (2021). A single-cell map of intratumoral changes during anti-PD1 treatment of patients with breast cancer. *Nat. Med.* **27**, 820–832.

Basso, K., and Dalla-Favera, R. (2012). Roles of BCL6 in normal and transformed germinal center B cells. *Immunol. Rev.* **247**, 172–183.

Bianchini, G., Balko, J.M., Mayer, I.A., Sanders, M.E., and Gianni, L. (2016). Triple-negative breast cancer: challenges and opportunities of a heterogeneous disease. *Nat. Rev. Clin. Oncol.* **13**, 674–690.

Blank, C.U., Haining, W.N., Held, W., Hogan, P.G., Kallies, A., Lugli, E., Lynn, R.C., Philip, M., Rao, A., Restifo, N.P., et al. (2019). Defining 'T cell exhaustion'. *Nat. Rev. Immunol.* **19**, 665–674.

Brescia, P., Schneider, C., Holmes, A.B., Shen, Q., Hussein, S., Pasqualucci, L., Basso, K., and Dalla-Favera, R. (2018). MEF2B instructs germinal center development and acts as an oncogene in B cell lymphomagenesis. *Cancer Cell* **34**, 453–465.e9.

Browaeys, R., Saelens, W., and Saeys, Y. (2020). NicheNet: modeling intercellular communication by linking ligands to target genes. *Nat. Methods* **17**, 159–162.

Cabrita, R., Lauss, M., Sanna, A., Donia, M., Larsen, M.S., Mitra, S., Johansson, I., Phung, B., Harbst, K., Vallon-Christersson, J., et al. (2020). Tertiary lymphoid structures improve immunotherapy and survival in melanoma. *Nature* **577**, 561.

Cancel, J.C., Crozat, K., Dalod, M., and Mattiuz, R. (2019). Are conventional type 1 dendritic cells critical for protective antitumor immunity and how? *Front. Immunol.* **10**, 9.

Cheng, S., Li, Z., Gao, R., Xing, B., Gao, Y., Yang, Y., Qin, S., Zhang, L., Ouyang, H., Du, P., et al. (2021). A pan-cancer single-cell transcriptional atlas of tumor infiltrating myeloid cells. *Cell* **184**, 792–809.e23.

Cusanovich, D.A., Daza, R., Adey, A., Pliner, H.A., Christiansen, L., Gunderson, K.L., Steemers, F.J., Trapnell, C., and Shendure, J. (2015). Multiplex single cell profiling of chromatin accessibility by combinatorial cellular indexing. *Science* **348**, 910–914.

DePristo, M.A., Banks, E., Poplin, R., Garimella, K.V., Maguire, J.R., Hartl, C., Philippakis, A.A., del Angel, G., Rivas, M.A., Hanna, M., et al. (2011). A framework for variation discovery and genotyping using next-generation DNA sequencing data. *Nat. Genet.* **43**, 491–498.

Dirix, L.Y., Takacs, I., Jerusalem, G., Nikolinakos, P., Arkenau, H.T., Forero-Torres, A., Boccia, R., Lippman, M.E., Somer, R., Smakal, M., et al. (2018). Avelumab, an anti-PD-L1 antibody, in patients with locally advanced or metastatic breast cancer: a phase 1b JAVELIN Solid Tumor study. *Breast Cancer Res. Treat.* **167**, 671–686.

Duhen, T., Duhen, R., Montler, R., Moses, J., Moudgil, T., de Miranda, N.F., Goodall, C.P., Blair, T.C., Fox, B.A., McDermott, J.E., et al. (2018). Co-expression of CD39 and CD103 identifies tumor-reactive CD8 T cells in human solid tumors. *Nat. Commun.* **9**, 1–13.

Eisenhauer, E.A., Therasse, P., Bogaerts, J., Schwartz, L.H., Sargent, D., Ford, R., Dancey, J., Arbuck, S., Gwyther, S., Mooney, M., et al. (2009). New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur. J. Cancer* **45**, 228–247.

Ferris, S.T., Durai, V., Wu, R., Theisen, D.J., Ward, J.P., Bern, M.D., Davidson, J.T., Bagadia, P., Liu, T.T., Briseno, C.G., et al. (2020). cDC1 prime and are licensed by CD4(+)T cells to induce anti-tumour immunity. *Nature* **584**, 624.

Gajewski, T.F., Schreiber, H., and Fu, Y.X. (2013). Innate and adaptive immune cells in the tumor microenvironment. *Nat. Immunol.* **14**, 1014–1022.

Guo, X., Zhang, Y., Zheng, L., Zheng, C., Song, J., Zhang, Q., Kang, B., Liu, Z., Jin, L., Xing, R., et al. (2018). Global characterization of T cells in non-small-cell lung cancer by single-cell sequencing. *Nat. Med.* **24**, 978–985.

Hafemeister, C., and Satija, R. (2019). Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Genome Biol.* **20**, 1–15.

Hargadon, K.M., Johnson, C.E., and Williams, C.J. (2018). Immune checkpoint blockade therapy for cancer: an overview of FDA-approved immune checkpoint inhibitors. *Int. Immunopharmacol.* **62**, 29–39.

Hegde, P.S., and Chen, D.S. (2020). Top 10 challenges in cancer immunotherapy. *Immunity* **52**, 17–35.

Helmink, B.A., Reddy, S.M., Gao, J.J., Zhang, S.J., Basar, R., Thakur, R., Yizhak, K., Sade-Feldman, M., Blando, J., Han, G.C., et al. (2020). B cells and tertiary lymphoid structures promote immunotherapy response. *Nature* **577**, 549.

Hollern, D.P., Xu, N., Thennavan, A., Glodowski, C., Garcia-Recio, S., Mott, K.R., He, X.P., Garay, J.P., Carey-Ewend, K., Marron, D., et al. (2019). B cells and T follicular helper cells mediate response to checkpoint inhibitors in high mutation burden mouse models of breast cancer. *Cell* **179**, 1191.

House, I.G., Savas, P., Lai, J., Chen, A.X.Y., Oliver, A.J., Teo, Z.L., Todd, K.L., Henderson, M.A., Giuffrida, L., Petley, E.V., et al. (2020). Macrophage-derived CXCL9 and CXCL10 are required for antitumor immune responses following immune checkpoint blockade. *Clin. Cancer Res.* **26**, 487–504.

Koboldt, D.C., Fulton, R.S., McLellan, M.D., Schmidt, H., Kalicki-Verizer, J., McMichael, J.F., Fulton, L.L., Dooling, D.J., Ding, L., Mardis, E.R., et al.

- (2012). Comprehensive molecular portraits of human breast tumours. *Nature* 490, 61–70.
- Krieg, C., Nowicka, M., Guglietta, S., Schindler, S., Hartmann, F.J., Weber, L.M., Dummer, R., Robinson, M.D., Levesque, M.P., and Becher, B. (2018). High-dimensional single-cell analysis predicts response to anti-PD-1 immunotherapy. *Nat. Med.* 24, 144–153.
- Lavin, Y., Kobayashi, S., Leader, A., Amir, E.D., Elefant, N., Bigenwald, C., Remark, R., Sweeney, R., Becker, C.D., Levine, J.H., et al. (2017). Innate immune landscape in early lung adenocarcinoma by paired single-cell analyses. *Cell* 169, 750–765.
- Li, H., and Durbin, R. (2009). Fast and accurate short read alignment with Burrows–Wheeler transform. *Bioinformatics* 25, 1754–1760.
- Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., Durbin, R., and Genome Project Data Processing, S. (2009). The sequence alignment/map format and SAMtools. *Bioinformatics* 25, 2078–2079.
- Li, H., van der Leun, A.M., Yofe, I., Lubling, Y., Gelbard-Solodkin, D., van Akkooi, A.C.J., van den Braber, M., Rozeman, E.A., Haanen, J., Blank, C.U., et al. (2019). Dysfunctional CD8 T cells form a proliferative, dynamically regulated compartment within human melanoma. *Cell* 176, 775–789.e18.
- Litchfield, K.R., Reading, J., McGranahan, N., Quezada, S., and Swanton, C. (2020). Meta-analysis of tumour and T cell intrinsic mechanisms of sensitization to checkpoint inhibition. *Ann. Oncol.* 31, S1092.
- Maier, B., Leader, A.M., Chen, S.T., Tung, N., Chang, C., LeBerichel, J., Chudnovskiy, A., Maskey, S., Walker, L., Finnigan, J.P., et al. (2020). A conserved dendritic-cell regulatory program limits antitumour immunity. *Nature* 580, 257.
- Miles, D., Gligorov, J., André, F., Cameron, D., Schneeweiss, A., Barrios, C., Xu, B., Wardley, A., Kaen, D., Andrade, L., et al. (2020). Primary results from IMpassion131, a double-blind placebo-controlled randomized phase III trial of first-line paclitaxel atezolizumab for unresectable locally advanced/metastatic triple-negative breast cancer. *Ann. Oncol.* 32, 994–1004.
- Molgora, M., Esaulova, E., Vermi, W., Hou, J.C., Chen, Y., Luo, J.Q., Brioschi, S., Bugatti, M., Omodei, A.S., Ricci, B., et al. (2020). TREM2 modulation re-models the tumor myeloid landscape enhancing anti-PD-1 immunotherapy. *Cell* 182, 886.
- Nanda, R., Chow, L.Q., Dees, E.C., Berger, R., Gupta, S., Geva, R., Pusztai, L., Pathiraja, K., Aktan, G., Cheng, J.D., et al. (2016). Pembrolizumab in patients with advanced triple-negative breast cancer: phase Ib KEYNOTE-012 study. *J. Clin. Oncol.* 34, 2460–2467.
- Olines, M.J., Kotliarov, Y., Biancotto, A., Cheung, F., Chen, J., Shi, R., Zhou, H., Wang, E., Tsang, J.S., Nussenblatt, R., et al. (2016). Effects of systemically administered hydrocortisone on the human immunome. *Sci. Rep.* 6, 23002.
- Pardoll, D.M. (2012). The blockade of immune checkpoints in cancer immunotherapy. *Nat. Rev. Cancer* 12, 252–264.
- Peperzak, V., Veraar, E.A., Xiao, Y., Babala, N., Thiadens, K., Brugmans, M., and Borst, J. (2013). CD8+ T cells produce the chemokine CXCL10 in response to CD27/CD70 costimulation to promote generation of the CD8+ effector T cell pool. *J. Immunol.* 191, 3025–3036.
- Petitprez, F., de Reynies, A., Keung, E.Z., Chen, T.W.W., Sun, C.M., Calderaro, J., Jeng, Y.M., Hsiao, L.P., Lacroix, L., Bougouin, A., et al. (2020). B cells are associated with survival and immunotherapy response in sarcoma. *Nature* 577, 556.
- Philip, M., Fairchild, L., Sun, L., Horste, E.L., Camara, S., Shakiba, M., Scott, A.C., Viale, A., Lauer, P., Merghoub, T., et al. (2017). Chromatin states define tumour-specific T cell dysfunction and reprogramming. *Nature* 545, 452–456.
- Polanski, K., Young, M.D., Miao, Z.C., Meyer, K.B., Teichmann, S.A., and Park, J.E. (2020). BBKNN: fast batch alignment of single cell transcriptomes. *Bioinformatics* 36, 964–965.
- Qu, Y., Wen, J., Thomas, G., Yang, W., Prior, W., He, W., Sundar, P., Wang, X., Potluri, S., and Salek-Ardakani, S. (2020). Baseline frequency of inflammatory Cxcl9-expressing tumor-associated macrophages predicts response to Avelumab treatment. *Cell Rep.* 32, 107873.
- R Core Team (2008). The R project for statistical computing (Vienna, Austria: T Team), Available online at <https://www.R-project.org/>.
- Ribas, A., and Wolchok, J.D. (2018). Cancer immunotherapy using checkpoint blockade. *Science* 359, 1350.
- Ritchie, M.E., Phipson, B., Wu, D., Hu, Y.F., Law, C.W., Shi, W., and Smyth, G.K. (2015). Limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* 43, e47.
- Salgado, R., Denkert, C., Demaria, S., Sirtaine, N., Klauschen, F., Pruneri, G., Wienert, S., Van den Eynden, G., Baehner, F.L., Penault-Llorca, F., et al. (2015). The evaluation of tumor-infiltrating lymphocytes (TILs) in breast cancer: recommendations by an International TILs Working Group 2014. *Ann. Oncol.* 26, 259–271.
- Saunders, C.T., Wong, W.S.W., Swamy, S., Becq, J., Murray, L.J., and Cheetham, R.K. (2012). Strelka: accurate somatic small-variant calling from sequenced tumor-normal sample pairs. *Bioinformatics* 28, 1811–1817.
- Savas, P., Virassamy, B., Ye, C., Salim, A., Mintoff, C.P., Caramia, F., Salgado, R., Byrne, D.J., Teo, Z.L., Dushyanthen, S., et al. (2018). Single-cell profiling of breast cancer T cells reveals a tissue-resident memory subset associated with improved prognosis. *Nat. Med.* 24, 986–993.
- Schmid, P., Adams, S., Rugo, H.S., Schneeweiss, A., Barrios, C.H., Iwata, H., Dieras, V., Hegg, R., Im, S.A., Wright, G.S., et al. (2018). Atezolizumab and nab-paclitaxel in advanced triple-negative breast cancer. *New Engl. J. Med.* 379, 2108–2121.
- Sen, D.R., Kaminski, J., Barnitz, R.A., Kurachi, M., Gerdemann, U., Yates, K.B., Tsao, H.W., Godec, J., LaFleur, M.W., Brown, F.D., et al. (2016). The epigenetic landscape of T cell exhaustion. *Science* 354, 1165–1169.
- Sharma, P., and Allison, J.P. (2015). Immune checkpoint targeting in cancer therapy: toward combination strategies with curative potential. *Cell* 161, 205–214.
- Simoni, Y., Becht, E., Fehlings, M., Loh, C.Y., Koo, S.L., Teng, K.W.W., Yeong, J.P.S., Nahar, R., Zhang, T., Kared, H., et al. (2018). Bystander CD8(+) T cells are abundant and phenotypically distinct in human tumour infiltrates. *Nature* 557, 575.
- Siu, L.L., Ivy, P., Dixon, E.L., Gravell, A.E., Reeves, S.A., and Rosner, G.L. (2017). Challenges and opportunities in adapting clinical trial design for immunotherapies. *Clin. Cancer Res.* 23, 4950–4958.
- Spitzer, M.H., Carmi, Y., Reticker-Flynn, N.E., Kwek, S.S., Madhiredy, D., Martins, M.M., Gherardini, P.F., Prestwood, T.R., Chabon, J., Bendall, S.C., et al. (2017). Systemic immunity is required for effective cancer immunotherapy. *Cell* 168, 487–502.e15.
- Srivatsan, S.R., McFaline-Figueroa, J.L., Ramani, V., Saunders, L., Cao, J.Y., Packer, J., Pliner, H.A., Jackson, D.L., Daza, R.M., Christiansen, L., et al. (2020). Massively multiplex chemical transcriptomics at single-cell resolution. *Science* 367, 45.
- Stuart, T., Butler, A., Hoffman, P., Hafemeister, C., Papalexi, E., Mauck, W.M., 3rd, Hao, Y., Stoeckius, M., Smibert, P., and Satija, R. (2019). Comprehensive integration of single-cell data. *Cell* 177, 1888–1902.e21.
- Tirosh, I., Izar, B., Prakadan, S.M., Wadsworth, M.H., 2nd, Treacy, D., Trombetta, J.J., Rotem, A., Rodman, C., Lian, C., Murphy, G., et al. (2016). Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science* 352, 189–196.
- Traag, V.A., Waltman, L., and van Eck, N.J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* 9, 5233.
- Van Rossum, G., and Drake, F.L. (2009). Python 3 Reference Manual (CreateSpace).
- Vogelstein, B., Papadopoulos, N., Velculescu, V.E., Zhou, S.B., Diaz, L.A., and Kinzler, K.W. (2013). Cancer genome landscapes. *Science* 339, 1546–1558.
- Wang, K., Li, M.Y., and Hakonarson, H. (2010). ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. *Nucleic Acids Res.* 38, e164.
- Wherry, E.J. (2011). T cell exhaustion. *Nat. Immunol.* 12, 492–499.
- Wolf, F.A., Angerer, P., and Theis, F.J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* 19, 1–5.

Wolock, S.L., Lopez, R., and Klein, A.M. (2019). Scrublet: computational identification of cell doublets in single-cell transcriptomic data. *Cell Syst.* 8, 281–291.e289.

Wu, T.D., and Nacu, S. (2010). Fast and SNP-tolerant detection of complex variants and splicing in short reads. *Bioinformatics* 26, 873–881.

Xu, W., Zhao, X.H., Wang, X.S., Feng, H., Gou, M.T., Jin, W., Wang, X.H., Liu, X.D., and Dong, C. (2019). The transcription factor Tox2 drives T follicular helper cell development via regulating chromatin accessibility. *Immunity* 51, 826.

Yost, K.E., Satpathy, A.T., Wells, D.K., Qi, Y., Wang, C., Kageyama, R., McNamara, K.L., Granja, J.M., Sarin, K.Y., Brown, R.A., et al. (2019). Clonal replacement of tumor-specific T cells following PD-1 blockade. *Nat. Med.* 25, 1251–1259.

Zhang, L., Li, Z.Y., Skrzypczynska, K.M., Fang, Q., Zhang, W., O'Brien, S.A., He, Y., Wang, L., Zhang, Q.M., Kim, A., et al. (2020). Single-cell analyses inform mechanisms of myeloid-targeted therapies in colon cancer. *Cell* 181, 442.

Zhang, L., Yu, X., Zheng, L., Zhang, Y., Li, Y., Fang, Q., Gao, R., Kang, B., Zhang, Q., Huang, J.Y., et al. (2018). Lineage tracking reveals dynamic relationships of T cells in colorectal cancer. *Nature* 564, 268–272.

Zhang, Q.M., He, Y., Luo, N., Patel, S.J., Han, Y.J., Gao, R.R., Modak, M., Carotta, S., Haslinger, C., Kind, D., et al. (2019). Landscape and dynamics of single immune cells in hepatocellular carcinoma. *Cell* 179, 829.

Zheng, C., Zheng, L., Yoo, J.K., Guo, H., Zhang, Y., Guo, X., Kang, B., Hu, R., Huang, J.Y., Zhang, Q., et al. (2017a). Landscape of infiltrating T cells in liver cancer revealed by single-cell sequencing. *Cell* 169, 1342–1356.e16.

Zheng, G.X.Y., Terry, J.M., Belgrader, P., Ryvkin, P., Bent, Z.W., Wilson, R., Ziraldo, S.B., Wheeler, T.D., McDermott, G.P., Zhu, J.J., et al. (2017b). Massively parallel digital transcriptional profiling of single cells. *Nat. Commun.* 8, 1–12.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT OR RESOURCES	SOURCE	IDENTIFIER
Antibodies		
Rabbit monoclonal anti-PD-L1	Roche	Clone SP142
Mouse anti-human CD4	Zsbio	Clone UMAB64
Mouse anti-human CD8	CST	Clone C8/144B; RRID: AB_2799781
Mouse anti-human PD-1	CST	Clone EH33; RRID: AB_2728836
Rabbit anti-human CXCL13	Abcam	Clone EPR23400-92
Rabbit anti-human CD19	CST	Clone D4V4B; RRID: AB_2800152
Rabbit anti-human CD68	Biolyx	Clone BP6036
Rabbit anti-human MMP9	CST	Clone D6O3H; RRID: AB_2798289
PANO 7-plex IHC kit	Panovue	Cat#0004100100
Library preparation and sequencing reagents		
Chromium™ Single Cell 5' Library Construction Kit	10x Genomics	Cat#1000020
Chromium™ Next GEM Single Cell 5' Library and Gel Bead Kit v1.1	10x Genomics	Cat#1000165
Chromium™ Single Cell A Chip Kit	10x Genomics	Cat#1000213
Chromium™ Single Cell V(D)J Enrichment Kit, Human T Cell	10x Genomics	Cat#1000005
Chromium™ Next GEM Single Cell ATAC Library & Gel Bead Kit v1.1	10x Genomics	Cat#1000175
Chromium™ Next GEM Chip H Single Cell Kit	10x Genomics	Cat#1000161
AllPrep DNA/RNA Mini Kit	QIAGEN	Cat#80204
Qubit dsDNA BR Assay Kit	Thermo Fisher Scientific	Cat# Q32850
SureSelectXT Human All Exon V5	Agilent	Cat#5190-6210
KAPA Hyper Prep kit	KAPA	KK8504
AllPrep DNA/RNA Mini Kit	QIAGEN	Cat#80204
Qubit RNA HS assay	Thermo Fisher Scientific	Cat# Q32855
TruSeq RNA Access Library Prep Kit	Illumina	Cat# RS-301-2001
Deposited data		
Processed single cell RNA-seq and ATAC-seq data in human TNBC patients	This paper	GEO:GSE169246
Processed single cell RNA-seq data in TNBC mouse models	Hollern et al. (2019)	GEO:GSE136206
Processed single cell RNA-seq data in human BCC patients	Yost et al. (2019)	GEO:GSE123814
TCGA LUAD, SKCM and BRCA datasets	XENA data portal	http://xena.ucsc.edu/
Softwares and algorithms		
Cell Ranger-3.1.0	10 X Genomics	http://10xgenomics.com/
Python-3.6.1	Van Rossum and Drake (2009)	https://www.python.org/
Scanpy- 1.6.0	Wolf et al. (2018)	https://github.com/theislab/Scanpy
R-3.6.0	R Core Team (2008)	https://www.r-project.org/
Seurat-3.2.2	Stuart et al. (2019)	https://satijalab.org/seurat/
NicheNet -0.1.0	Browaeys et al. (2020)	https://github.com/saeyslab/nichenetr
STARTRAC	Zhang et al., 2018	https://github.com/Japrin/STARTRAC
GSNAP	Wu and Nacu (2010)	http://research-pub.gene.com/gmap/

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REAGENT OR RESOURCES	SOURCE	IDENTIFIER
BWA-0.7.12	Li et al. (2009)	https://github.com/lh3/bwa
GATK-4.0.5.2	DePristo et al. (2011)	https://software.broadinstitute.org/gatk/
Strelka-1.0.14	Saunders et al. (2012)	ftp://strelka@ftp.illumina.com/
Limma-3.42.0	Ritchie et al. (2015)	http://bioinf.wehi.edu.au/limma
Samtools-0.1.19	Li et al. (2009)	http://www.htslib.org/
ANNOVAR	Wang et al. (2010)	https://doc-openbio.readthedocs.io/projects/annovar/en/latest/
Other		
Interactive portal for immune cell dynamics in human TNBC	This paper	http://TNBC_PD-L1.cancer-pku.cn

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Zemin Zhang (zemin@pku.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

Processed single cell RNA-seq and ATAC-seq data of this study can be obtained from Gene Expression Omnibus (GEO) with an accession number of GSE169246. Treatment-related single cell RNA-seq datasets of TNBC mouse models (Hollern et al., 2019) and human BCC patients (Yost et al., 2019) are available from GEO with an accession number of GSE136206 and GSE123814, respectively. Bulk RNA-seq datasets of TCGA LUAD, SKCM and BRCA can be obtained from XENA data portal (<http://xena.ucsc.edu/>). The raw FASTQ files in this study will be provided for scientific research upon request complying with the law due to human patient privacy concerns. Code used for all processing and analysis is available upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human specimens

Eligible TNBC patients were identified based on prospective chart review through the DTHealth TrakCare System at National Cancer Center, Beijing, China. Patients enrolled in this study were those who have histologically confirmed advanced TNBC, no prior chemotherapy or targeted systemic therapy, and received paclitaxel monotherapy or paclitaxel plus atezolizumab as first-line treatment for advanced TNBC. Patients receive treatment according to clinical routine, with paclitaxel 90 mg/m² on Days 1, 8, and 15 of every 28-day cycle, and atezolizumab 840 mg on Days 1 and 15 of every 28-day cycle (NCC Protocol #1850). This study was approved by the Ethical Committee of National Cancer Center/Cancer Hospital, Chinese Academy of Medical Sciences (No. 18-216/1794) and complied with all relevant ethical regulations. All patients provided written informed consent for the collection of tissue and blood samples for research and genomic profiling.

A total of 22 TNBC patients were included in this study, half of which received chemotherapy (paclitaxel monotherapy), while the other half received the combination therapy of PD-L1 blockade and chemotherapy (atezolizumab plus paclitaxel). All patients in this study were diagnosed with advanced TNBC, and most patients had metastatic disease. Indeed, some patients were locally advanced patients receiving neoadjuvant therapy (Table S1). Patients were female in gender and their ages ranged from 32-64 years with a median value of 49. Of note, we collected breast tumor samples when available, while collected metastatic tumor samples instead if patients had already gone through breast tumor surgical removal. Specifically, for patients in the chemotherapy group, breast tumors were collected from 6 patients, and metastatic tumors were collected from 3 patients. By contrast, metastatic tumors were collected from 9 patients in the combination therapy group, since these patients had already gone through breast tumor surgical removal, and only metastatic tumors could be obtained. For the 11 patients treated with paclitaxel plus atezolizumab, 5 patients harbored PD-L1⁺ tumors, among which 3 were responders and 2 were non-responders, while 6 patients harbored PD-L1⁻ tumors, among which 1 was responder and 5 was non-responders (Figure S1A; Table S1). We also evaluated TIL levels in TNBC tumors, and obtained such data for 8 patients in the combination therapy group. In these patients, 4 patients showed high levels of TILs, 3 of which were responders while 1 was non-responder. The other 4 patients showed low levels of TILs and were non-responders (Figure S1B; Table S1). Detailed information of individual patients in this study has been provided in Table S1.

Paired biopsies samples of metastatic or primary tumor tissue and peripheral blood were collected at baseline of < 2 weeks prior to treatment, 4 weeks after treatment initiation, as well as disease progression. All patients except P003 and P028 had paired peripheral blood samples before and after treatment, among which 7 patients had blood samples after disease progression. Eighteen patients had tumor biopsy samples, half in the combination therapy group while the other half in the chemotherapy group, and 2 patients had tumor samples after diseases progression. Clinical tumor biopsy samples of P008 and P024 failed in quality control process before library preparation (Table S1).

Clinical responses were evaluated based on radiologic assessments of tumor sizes by the investigator every 8 weeks (two treatment cycles) according to the Response Evaluation Criteria in Solid Tumors (RECIST, version 1.1) (Eisenhauer et al., 2009). Ultrasound was used to evaluate the breast tumors, and enhanced computerized tomography (CT) was used to evaluate the metastatic tumors in chest or abdomen. These assessment approaches are in compliance with international standards. Of note, the final response classification in our study was based on the patient responses of the 8th week after treatment initiation, which was the first response assessment since the biopsy samples were obtained. Both the combination therapy and chemotherapy groups received a median of 6.0 cycles of treatment. It should be noted that multiple other parameters could also be used as criteria for patient response evaluation, including the best overall response during the entire treatment process, and the duration of response or stable disease (or alternatively time to progression, TTP). Different parameters show certain inconsistency, which is common in clinical practice. In our study, we used the responses evaluated at the 8th week after treatment initiation as the final response classification in order to directly connect the dynamics of immune cell compositions with patient responses following the first two-cycle treatment. We classified each of the 22 patients into responder/regression (R; n=9, including CR/PR samples) or non-responder/progression (NR; n = 12, including SD/PD samples) (Table S1). Specifically, we obtained 4 responders (PR) and 7 non-responders (SD) for the combination therapy group, while 5 responders (PR), 5 non-responders (3 SD and 2 PD), and 1 not available for the chemotherapy group (Table S1).

METHOD DETAILS

Sample collection and processing

Peripheral blood mononuclear cells (PBMCs) were isolated with HISTOPAQUE-1077 (Sigma-Aldrich) solution as previously described (Zhang et al., 2018; Zheng et al., 2017a). In brief, 3 ml fresh peripheral blood was collected at baseline, 4 weeks after treatment initiation and disease progression in EDTA anticoagulant tubes and subsequently layered onto HISTOPAQUE-1077. After centrifugation, lymphocyte cells remained at the plasma-HISTOPAQUE-1077 interface and were carefully transferred to a new tube and washed twice with 1x phosphate-buffered saline (PBS, Invitrogen). After the supernatant was removed, the pelleted cells were suspended in red blood cell lysis buffer (Solarbio) and incubated on ice for 2 min to lyse red blood cells. These lymphocytes were re-suspended in sorting buffer PBS supplemented with 1% fetal bovine serum (FBS, Gibco).

For tumor biopsy samples, pre- and post-treatment samples were site-matched and were obtained from the same primary or metastatic tumor before and after 4 weeks of treatment. Each tumor was punctured by two or three needles, using 16-gauge needles for breast tumors or chest wall metastatic tumors, and 18-gauge needles for metastatic tumors in lymph nodes, liver or lung tissues. All sample collection procedures complied with the regular routine in clinical practice. Tumor biopsy samples were first stored in RNA-later RNA stabilization reagent (QIAGEN) after collection and kept on ice to avoid RNA degradation. Biopsy samples from each tumor at each time point were then combined before tissue disassociation. Samples were then collected in the RPMI-1640 medium (Gibco), and enzymatically digested with gentle MACS Tumor Dissociation Kit (Miltenyi Biotec) for 60 min on a rotor at 37 °C according to the manufacturer's protocol. The dissociated cells were next passed through a 40- μ m cell-strainer (BD) in the RPMI-1640 medium (Invitrogen) with 10% FBS until uniform cell suspensions were obtained. Subsequently, the suspended cells were passed through cell strainers and centrifuged at 400 g for 10 min. Red blood cells were removed via the same procedure described above. After washing twice with 1x PBS (Invitrogen), the cell pellets were re-suspended in sorting buffer (PBS supplemented with 1% FBS).

Single-cell RNA-seq library and TCR-seq library preparation and sequencing

Single-cell RNA-seq libraries were prepared using the Chromium Next GEM Single Cell 5' Kit v2 from 10x Genomics, following the manufacturer's instructions. In brief, single cells were washed once with PBS containing 0.04% bovine serum albumin (BSA) and re-suspended in PBS containing 0.04% BSA to a final concentration of 500~1,200 cells/ μ l as determined by Rigel S2 cell counter (Countstar). ~10,000 cells were captured in droplets to generate nanoliter-scale Gel bead in EMulsion (GEMs). GEMs were then reverse transcribed in a C1000 Touch Thermal Cycler (Bio-Rad) programmed at 53°C for 45 min, 85°C for 5 min, and held at 4°C. After reverse transcription and cell barcoding, emulsions were broken and cDNA was isolated and purified with Cleanup Mix containing DynaBeads and SPRIselect reagent (Thermo Fisher Scientific), followed by PCR amplification. Amplified cDNA was then used for both 5' gene expression library construction and TCR V(D)J targeted enrichment performed with the Chromium Single Cell V(D)J Enrichment Kit, Human T cell (10x Genomics), followed by V(D)J library construction. For RNA-seq library construction, amplified cDNA was fragmented and end-repaired, double-sided size-selected, PCR-amplified with sample indexing primers, and double-sided size-selected. For TCR-seq library construction, TCR transcripts were enriched from amplified cDNA by PCR. Subsequently, enriched PCR product was fragmented and end-repaired, size-selected, PCR-amplified with sample-indexing primers, and size-selected. Libraries prepared according to the manufacturer's user guide were then purified and profiled for quality assessment. Single-cell RNA and TCR V(D)J libraries were sequenced by an Illumina HiSeq X-Ten sequencer with 150 bp paired-end reads.

Single-cell ATAC-seq library preparation and sequencing

Single-cell ATAC-seq libraries were prepared according to the Chromium Single Cell ATAC Library & Gel Bead Kit v1 and Chip Kit from 10x Genomics, following the manufacturer's instructions. Briefly, the single cell nuclei were suspended in 1× diluted nuclei buffer (10x Genomics), and then nuclei concentrations were adjusted to the desired capture number after counting (100–800 cells/μl). The re-suspended nuclei were next incubated at 37°C for 60 min with 10 μl of transposition mix composed of 7 μl ATAC Buffer, and 3 μl ATAC Enzyme. The transposed nuclei were then mixed with a master mixture composed of Barcoding Reagent, Reducing Agent B and Barcoding Enzyme, which were loaded onto a Chromium Chip to generate nanoliter-scale GEMs. GEMs were then collected and linearly amplified in a C1000 Touch Thermal Cycler. Emulsions were coalesced using the Recovery Agent, and then purified with Cleanup Mix containing DynaBeads and SPRIselect reagent (Thermo Fisher Scientific), followed by PCR amplification. Subsequently, the sequencing library was purified with SPRIselect beads, quantified and sequenced by an Illumina NovaSeq 6000 with 50-bp paired-end reads.

Bulk DNA and RNA isolation and sequencing

The bulk DNA and RNA isolation, library construction and sequencing were performed at Mingma Technologies Co., Ltd at Shanghai, China. Genomic DNA of peripheral blood and 10,000~20,000 single cells derived from tumor biopsies of TNBC patients was extracted using the QIAamp DNA Blood Mini Kit (Cat#51106, QIAGEN) and AllPrep DNA/RNA Mini Kit (Cat#80204, QIAGEN) according to the manufacturer's protocol respectively, the concentrations of which were quantified with the Qubit dsDNA BR Assay Kit (Cat# Q32850, Thermo Fisher Scientific) and the qualities were evaluated with agarose gel electrophoresis. DNA of peripheral blood libraries were constructed using the SureSelectXT Human All Exon V5 (Cat#5190-6210, Agilent), while DNA of single cells derived from tumor biopsy libraries were prepared using KAPA Hyper Prep kit (KK8504, KAPA). Prepared libraries were sequenced by the Illumina HiSeq X-Ten sequencer with 150 bp paired-end reads.

RNA of pseudo-bulk samples consisting of 10,000~20,000 single cells derived from tumor biopsies of TNBC patients were extracted using AllPrep DNA/RNA Mini Kit (Cat#80204, QIAGEN) according to the manufacturer's protocol, the concentrations of which were quantified with Qubit RNA HS assay (Cat# Q32855, Thermo Fisher Scientific). Bulk RNA-seq libraries were constructed using TruSeq RNA Access Library Prep Kit (Cat# RS-301-2001, Illumina). About 10 ng high-quality RNA was fragmented, cDNA was generated using random priming, and then sequencing adapters were ligated. The coding regions of the transcriptome were then captured using sequence-specific probes to create the final libraries. Prepared libraries were quantified and sequenced by the Illumina HiSeq X-Ten sequencer with 150 bp paired-end reads.

SP142 PD-L1 immunohistochemistry

The protein levels of PD-L1 were evaluated in clinical samples of TNBC patients treated with the combination therapy using VENTANA PD-L1 (SP142) Assay, according to the manufacturer's instructions. Briefly, the rabbit monoclonal anti-PD-L1 clone SP142 was used to assess PD-L1 protein expression in the formalin-fixed, paraffin-embedded (FFPE) tissue sections of TNBC tumor samples, and visualized using OptiView DAB IHC Detection Kit followed by the OptiView Amplification Kit. For TNBC patients in the combination therapy group, metastatic tumor biopsy samples or their primary breast tumor specimens in FFPE archival were used to perform VENTANA PD-L1 Assay. Positive and negative controls were included (tonsil or prostate tissues). All slides were manually and independently reviewed by two pathologists and consensus evaluations were achieved. PD-L1 expression was assessed on tumor-infiltrating immune cells (IC), and a cutoff of 1% was used to discriminate PD-L1 positive or negative tumors.

TILs evaluation

We followed the international guidelines for the evaluation of TILs (Salgado et al., 2015). Stromal TILs were scored as a percentage of the stromal areas occupied by mononuclear inflammatory cells over total intratumoral stromal area on H&E-stained sections. Views that are representative of the total TIL levels were selected.

Multi-color immunohistochemistry

The specimens collected from Cancer Hospital, Chinese Academy of Medical Sciences within 30 min after the biopsy procedure were fixed in 10% formalin for 48 hr, and then dehydrated and embedded in paraffin following routine methods, as previously described (Zheng et al., 2017a). Briefly, Paraffin blocks were cut into 4 μm sections and adhered to a glass slide heated at 70 °C for 1 h, deparaffinized in xylene, and then rehydrated in 100%, 95%, 70% alcohol in sequence. To confirm the existence of B cells, CD8-CXCL13, and CD4-CXCL13 cells, multiplex immunofluorescence staining was performed using PANO 7-plex IHC Kit (Cat#0004100100, Panovue) according to the manufacturer's protocol. In brief, antigen was retrieved by Citric acid solution (pH 9.0, Panovue) and boiled in the oven for 15 min. After a pre-incubation with blocking buffer at room temperature for 10 min, the sections were incubated with rabbit anti-human CD19 (CST, clone D4V4B, 1:100), mouse anti-human CD4 (Zsbio, clone UMAB64, 1:50), and mouse anti-human CD8 (CST, clone C8/144B, 1:50) at room temperature for 30 min, mouse anti-human PD-1 (CST, clone EH33, 1:200) at 4°C overnight, and rabbit anti-human CXCL13 (Abcam, clone EPR23400-92, 1:50) at room temperature for 1 hr. A secondary horseradish peroxidase-conjugated antibody (Panovue) were added and incubated at room temperature for 10 min. Signal amplification was performed using TSA working solution diluted at 1:100 in 1× amplification diluent (Panovue) and incubated at room temperature for 10 min. Multi-color IHC were performed with the same protocols but different primary antibodies, including rabbit anti-human CD68 (Biolynx, clone BP6036, 1:200) and rabbit anti-human MMP9 (CST, clone D6O3H, 1:100),

to confirm the existence of immuno-stimulatory macrophages. Nuclei were stained with DAPI after all the antigens above have been labelled. The stained slides were scanned with Mantra System (PerkinElmer). For each slide, a total of 10-15 fields were taken based on their tissue sizes. The selected field was scanned to obtain multispectral images using the Mantra System. The quantitative analysis was performed using inForm image analysis software (Version 2.4, PerkinElmer, Waltham, Massachusetts, US).

Single-cell RNA-Seq data processing

The Cell Ranger toolkit (version 3.1.0) provided by 10x Genomics was applied to aggregate raw data, filter low-quality reads, align reads to human reference genome (GRCh38), assign cell barcodes, and generate the unique molecular identifier (UMI) matrix. A Python-based toolkit, Scanpy (version 2.3.4) (Wolf et al., 2018), was used for analyzing the single-cell RNA-Seq (scRNA-seq) data. Specifically, the raw UMI matrix was processed to filter out genes detected in less than 10 cells and cells with fewer than 200 genes. We further quantified the numbers of gene and UMI count for each cell, and kept high quality cells with thresholds of 600-120,000 UMIs, 400-8,000 genes and less than 10% mitochondrial gene counts, to ensure that most of the heterogeneous cell types were included for downstream analyses. Scrublet (Wolock et al., 2019) was then applied to each sequencing library to remove potential doublets with the expected doublet rate of 6%, and cells with doubletScore larger than 90% quantile were filtered out. The normalized expression matrix was calculated based on the raw UMI counts after normalizing total counts per cell (library size) and then scaled by 1e6 and logarithmically transformed.

Dimension reduction and unsupervised clustering for scRNA-seq data

Dimension reduction and unsupervised clustering were performed according to the standard workflow in Scanpy (Wolf et al., 2018). Briefly, top 4,000 highly variable genes (HVGs) were detected using the dispersion-based methods by setting flavor='cell_ranger', where the normalized dispersion is obtained by scaling with the mean and standard deviation of the dispersions for genes falling into a given bin (Zheng et al., 2017b). Unwanted sources of variation including total counts, percentages of mitochondrial gene counts and heat shock protein (HSP) related gene counts were further regressed out from the normalized expression matrices. Principal component analysis (PCA) was performed on the variable gene matrix to reduce noise, and top 50 components were used for downstream analyses. To correct the batch effects from different patients, we applied BBKNN to generate a batch-balanced k nearest neighbor (KNN) graph, which identifies the top neighbors of each cell in each batch separately instead of the entire cell pool (Polanski et al., 2020). Then, Leiden algorithm was applied on such nearest neighbor graphs to detect communities and find cell clusters (Traag et al., 2019). Of note, the same principal components were also used for non-linear dimension reduction to generate the Uniform Manifold Approximation and Projection (UMAP) or t-Distributed Stochastic Neighbor Embedding (t-SNE) projections for visualization.

After the first-round of unsupervised clustering, we annotated each cell cluster according to canonical immune cell markers, and identified the major immune cell types including T cells, ILCs, B cells and myeloid cells. Notably, immune cell clusters derived from tumor tissue and peripheral blood were separated clearly, and thus, we performed a second-round of unsupervised clustering on each major immune cell compartment derived from tumor tissue or blood, respectively, to obtain the high-resolution map of immune cells in TNBC patients. The second-round clustering procedure was the same as the first-round clustering, both of which started from unfiltered expression matrix, and then identified HVGs, calculated PCA matrix, corrected batch effects by BBKNN, detected cell clusters by Leiden algorithm and performed dimensionality reduction for visualization. All analyses were performed with Scanpy (Wolf et al., 2018). Marker genes were detected using the rank_genes_groups function with parameters of method='logreg' and n_genes=200. Of note, one subset of T cells derived from baseline tumor tissue of P002 was clearly separated from other cells, and contained a mixture of different T cell states, which might be dominated by its high expression of immunoglobulin G (IgG) related genes, and thus were excluded from the downstream analysis.

To detect DEGs in specific immune cell clusters, we normalized the raw count matrix by SCTransform method, which models the UMI counts using a regularized negative binomial model to remove the variation of sequencing depth, while adjusting the variance based on pooling information across genes with similar abundances (Hafemeister and Satija, 2019). The DEGs were then detected by limma package (Ritchie et al., 2015) with the normalized expression obtained via SCTransform.

Single-cell ATAC data processing

The single-cell ATAC-seq (scATAC-seq) data were aggregated and processed using Cell Ranger ATAC pipeline (version 1.1.0), which generated a count matrix for open chromatin regions in single cells after barcode processing, alignment, duplicate marking, peak calling and cell calling. Human reference genome version GRCh38 was used for alignment process. Downstream analyses were performed with the Signac workflow (version 1.1.0), an extension of Seurat designed for the analyses of single-cell chromatin data and its aggregation with scRNA-seq data (Stuart et al., 2019). In brief, the peak count matrix was first filtered to keep peaks detected in at least 10 cells and cells with at least 200 peaks. The genomic ranges and gene information were annotated for each peak. Then, a series of additional quality control metrics including nucleosome banding signal, transcriptional start site (TSS) enrichment score, total number of fragments in peaks, fraction of fragments in peaks and ratio of reads in genomic blacklist regions, were calculated for each single cell. High-quality cells with nucleosome banding signal less than 4, TSS enrichment score larger than 2, the number of fragments in peaks larger than 3,000 and less than 20,000, the fraction of fragments in peaks larger than 15%, and reads in genomic blacklist regions less than 5%, were kept for downstream analyses, resulting in 151,138 features across 30,758 cells, which contained 19,615 T cells (Table S6).

Dimension reduction and unsupervised clustering for scATAC-seq data

The latent semantic indexing (LSI) practice was then applied on the high-quality peak count matrix for downstream analyses (Cusanovich et al., 2015; Srivatsan et al., 2020). Specially, term frequency-inverse document frequency (TF-IDF) normalization was performed to correct differences in sequencing depths across cells and across peaks. Then the top 25% peaks were selected with the FindTopFeatures function followed by dimensional reduction with singular value decomposition (SVD) on the TF-IDF matrix. Of note, the first LSI component was removed for its correlation with sequencing depth determined by the DepthCor function, and the other top 30 LSI components were used to construct a shared nearest neighbor (SNN) graph by first determining the k-nearest neighbors of each cell with k=20, and then calculating the neighborhood overlap between each cell and its k-nearest neighbors with the FindNeighbors function. Finally, the smart local moving (SLM) algorithm was applied for modularity optimization and identified cell clusters with the FindClusters function. The same LSI components were also used for non-linear dimension reduction of UMAP and t-SNE projections to visualize cell clusters in a two-dimensional space.

To annotate cell clusters identified on scATAC-seq data, we generated a gene activity matrix by counting the fragments intersecting the gene body and promoter (defined as 2 kb upstream) regions based on the chromatin accessibility data with the GeneActivity function. The gene activity matrix was then normalized by the total feature counts, scaled with the median feature counts, and log-transformed. We next integrated scATAC-seq data with paired scRNA-seq data according to the integration procedure of Seurat (Stuart et al., 2019). In brief, we performed cross-modality integration with canonical correlation analysis (CCA) for dimension reduction and label transfer with projected LSI for anchor building, based on the gene activity matrix and the gene expression matrix. To obtain cell type annotations for scATAC-seq clusters, we calculated the confusion matrix based on their predicted labels, and annotated each scATAC-seq cluster as the cell type with the highest cell number. The cellular identities of each cluster were further confirmed with their marker gene activities. The signature peaks of each cluster were detected using the FindMarkers function, and annotated with the closest genes using the ClosestFeature function. The differentially accessible regions between cell clusters were detected by a differential accessibility (DA) test, in which logistic regression was utilized with the total number of fragments as a latent variable to mitigate the effect of differential sequencing depths. The pseudo-bulk accessibility tracks were visualized by the CoveragePlot function provided by Signac (Stuart et al., 2019).

TCR sequences assembly

Cell Ranger toolkit (version 3.1.0) provided by 10x Genomics was applied to align TCR-seq reads to the human reference genome (GRCh38) and assemble TCR sequences. The preliminary TCR sequences were filtered to keep those characterized with high confident, full-length, productive, and assigned with a valid cell barcode and an unambiguous chain type. Each cell was assigned with one pair of alpha and beta chains with the highest UMI counts. Cells with identical TCR pairs were defined as clonal cells and considered to originate from the same ancestry. In total, 55.72% (148,998/267,408) of T cells were assigned with one pair of TCR sequences, and 98,381 clonotypes were detected, with a median of 4,349 unique clonotypes in each patient (ranging from 1,612–10,727). The sizes of clonotypes ranged from 1–721 and 8,864 clonotypes comprised more than one cell (Table S3).

Bulk RNA-sequencing and exome-sequencing data processing

Bulk RNA-seq and exome-seq data were processed using the same pipeline as we previously described (Zhang et al., 2018; Zheng et al., 2017a). For bulk RNA-seq data, low-quality reads were filtered out according to the following criteria: (1) “N” bases accounting for more than 10% reads’ length, or (2) bases with quality < 5 accounting for more than 50% reads’ length, or (3) containing adaptor sequences. Reads passing the above filters were then aligned to ribosomal RNA sequences downloaded from the RFam database with the GSNAP alignment tool (Wu and Nacu, 2010). The unmapped reads were kept and mapped to the human reference genome (hg19) downloaded from UCSC. A gene model file named knownGene.txt downloaded from UCSC database was then used to define known exon-exon junctions with the adjustable parameters “-novelsplicing 1 -n 10 -i 1 -M 2” in addition to other default parameters. The read counts of each gene were obtained by retrieving reads overlapping with genes defined in the gene model file with R (version 3.5.0) function findOverlaps.

For bulk exome-seq data, low-quality reads were filtered out following the same procedure as bulk RNA-seq data, and high-quality reads were then mapped to the human reference genome (version b37 with decoy sequence added, download from Broad) with Burrows-Wheeler Aligner (BWA)-MEM (version 0.7.12) (Li and Durbin, 2009). Subsequently, PICARD (version 2.18.9) and GATK toolkit (version 3.8.1) were applied to sort bam file, mark duplicate reads, locally re-align reads around candidate insertions and deletions (INDELs), and re-calibrate base quality according to the suggested best practice. Strelka (version 1.0.14) (Saunders et al., 2012) was used to call somatic mutations including (single-nucleotide variant) SNV and INDEL, which were further manually inspected using the IGV tool and annotated with ANNOVAR (Wang et al., 2010). Candidate driver and hot-spot mutations were annotated according to the driver gene list previously proposed (Vogelstein et al., 2013) and COSMIC database.

Definition of predicative index and therapeutic index

To circumvent the limitation of sample sizes, we designed a Pi and a Ti to leverage the single-cell and clinical data in a continuous axis, and thus could detect the consistent patterns among immune cell compositions and their connections with treatment-induced clinical outcomes in all patients. The Pi was designed to determine whether a baseline cellular component is associated with the clinical response, as it measures the correlation between baseline cellular proportions and treatment-induced changes in tumor size. The baseline cellular proportions were calculated by the fractions of immune cell clusters in corresponding major immune compartments

that involve T cells, B cells, ILC and myeloid cells, in pre-treatment samples. The tumor size changes were calculated by the relative changes of tumor sizes between post- and pre-treatment samples. Specifically, the P_i was defined as $P_i = \frac{-slope}{|slope|} \times R^2$, where the *slope* and R^2 were obtained by a linear regression model $\text{lm}(y1 \sim x1)$ that estimates the correlation between the baseline cellular proportions ($x1$) of corresponding immune cell clusters with the tumor size changes ($y1$) in patients treated with different regimens. We utilized the 'geom_smooth' function with the parameter of 'method = lm' provided by R package ggplot2 to perform the linear regression model. A positive P_i represents that a higher baseline level of the corresponding immune cell cluster is associated with a better clinical response, while a negative P_i represents that a higher baseline level of the corresponding immune cell cluster is associated with a worse clinical response.

The T_i was designed to determine whether a dynamic change in cellular proportion was associated with the clinical response, as it measures the correlation between the treatment-induced cellular proportion changes and tumor size changes. The cellular proportion dynamics were calculated by the relative changes in cellular proportions for immune cell clusters in corresponding major immune compartments between post- and pre-treatment samples. The tumor size changes were calculated by the relative changes in tumor sizes between post- and pre-treatment samples. Specifically, the T_i was similarly defined as $T_i = \frac{-slope}{|slope|} \times R^2$, where the *slope* and R^2 were obtained by a linear regression model $\text{lm}(y2 \sim x2)$ that estimates the correlation between the cellular proportion changes ($x2$) of the corresponding immune cell cluster with the tumor size changes ($y2$) in patients treated with different regimens. A positive T_i represents that a higher level of increase in cellular proportion for the corresponding immune cell cluster is associated with a better clinical response, while a negative T_i represents that a higher level of increase in cellular proportion for the corresponding immune cell cluster is associated with a worse clinical response.

It should be noted that the P_i was based on the baseline levels of immune cell compositions and thus it was derived from patients with pre-treatment biopsy data. In contrast, the T_i was based on the relative dynamic changes of immune cell compositions between post- and pre-treatment samples, and thus it was derived only from patients with paired biopsy data.

Tissue enrichment and time-point distribution analysis

To quantify the enrichment of immune cell types across different tissues and time-points of sample collection, we compared the observed and expected cell numbers for each cluster in each tissue and time-point according to the following formula as we previously described (Guo et al., 2018; Zhang et al., 2018), $R_{o/e} = \frac{\text{Observed}}{\text{Expected}}$, where the expected cell numbers of immune cell clusters in a given tissue or time-point were calculated from the Chi-square test. We assumed that one cluster was enriched in a specific tissue or time-point if $R_{o/e} > 1$.

Definition of exhaustion score and effector memory score

To evaluate the transcriptional dynamics of potential tumor-reactive T cells of CD8-CXCL13 in responsive patients following the combination therapy, we defined exhaustion score and effector memory score based on a set of T cell exhaustion-related genes and a group of effector- or memory- related genes identified in our previous single-cell studies of tumor-infiltrating T cells (Zhang et al., 2018; Zheng et al., 2017a). The exhaustion score was defined as the average expression of *CXCL13*, *HAVCR2*, *PDCD1*, *TIGIT*, *LAG3*, *CTLA4*, *LAYN*, *RBPJ*, *VCAM1*, *GZMB*, *TOX* and *MYO7A*, while effector memory score as the mean expression of effector- or memory-related genes of *PRF1*, *IFNG*, *CCL4*, *HLA-DQA1*, *GZMK*, *GZMA*, *GZMH*, *CD44*, *DUSP2*, *KLRB1*, *KLRD1* and *CTSW*.

Survival analysis

The TCGA BRCA data were used to evaluate the prognostic performance of different immune cell clusters. The gene expression data and clinical information of BRCA patients were downloaded from UCSC Xena (<http://xena.ucsc.edu/>). We used the mean expression of signature genes for distinct cell clusters to evaluate the enrichment of corresponding immune cell types in BRCA patients. The signature genes were defined based on the combination of canonical marker genes of major immune cell types and cluster specific makers. Specifically, for distinct B cell clusters, we used the canonical marker genes of B cells including *CD79A* and *CD19* and cluster specific makers as provided in Table S2 to define their signatures, and defined Bn signature as *CD19*, *MS4A1*, *FCER2* and *TCL1A*, Bfoc signature as *CD19*, *MS4A1*, *NEIL1*, *RGS13*, *MEF2B* and *BCL6*, Bmem signature as *CD19*, *MS4A1*, *CD27* and *TNFRSF13B*, pB signature as *CD79A*, *MZB1*, *IGHG1* and *XBP1*. The expression specificities of these makers in B cell subsets have been shown in Figure S3A. The BRCA patients were classified into two groups according to the median value of mean expressions of a given signature. The Cox proportional hazards model implemented in R package 'survival' was used to perform survival analyses after correcting the effects of other clinical factors including tumor stage and patient age.

Correlations analysis

For single-cell based correlation analysis, we first calculated the baseline cellular proportions of each immune cell cluster by their fractions in corresponding major immune compartments, which involve T cell, B cell, ILC and myeloid cell compartments, in each patient before treatment. We then evaluated the baseline correlations of immune cell clusters by their Pearson correlation coefficient (PCC) calculated with their baseline cellular proportions in all patients or in patients treated with different regimens. Similarly, the post-treatment correlations of immune cell clusters were evaluated based on their cellular proportions in patients after treatment. For their correlated dynamic changes, we utilized the relative changes in cellular proportion following treatment, and calculated the PCC of immune cell clusters based on their cellular proportion changes in all patients or in patients treated with different regimens.

To confirm the connections of distinct immune cell clusters identified by our single-cell analyses, we utilized the signature genes of corresponding immune cell clusters and calculated their mean expressions in bulk RNA-seq data of TCGA BRCA. We used *LAMP3*, *FSCN1*, *IDO1*, *CCL19* and *EBI3* as mDC signature, *XCR1*, *CLEC9A*, *IDO1* and *CLNK* as cDC1 signature, and *CD4*, *CXCL13*, *IL6ST*, *TOX2*, *CD200* and *CXCR5* as CD4-CXCL13 signature. These marker genes were selected based on their specific expressions in each cell cluster by our single-cell data (Table S2).

Predictive performance evaluation for signature genes

To identify potential biomarkers applicable in clinical practice for anti-PD-L1 immunotherapy, we focused on immune cell subsets showing high predictive potentials and their signature genes. In addition to our Pi analyses that suggested the positive connections of CXCL13⁺ T cells and their highly correlated immune cell subsets with the favorable responses to the combination therapy, we further evaluated the enrichments of immune cell subsets in baseline responsive patients compared to non-responsive patients using Chi-Square test, after filtering immune cell clusters that contained less than 50 cells in baseline responsive samples. We used the ratio of observed to expected cell numbers ($R_{o/e}$) to estimate the enrichments of immune cell clusters in baseline responsive patients, and selected those with $R_{o/e} \geq 1.2$ and false discovery rate (FDR) < 0.01 as immune cell clusters showing significant enrichments in responders. We performed such enrichment analysis in both chemotherapy and the combination therapy groups. Immune cell clusters showing high Pi and significant enrichments in responders were defined as those with high predictive potentials.

Based on these two criteria, we found that CXCL13⁺ T cells and M ϕ -MMP9, which also exhibited significant enrichments in responders, showed consistent patterns in their connections with favorable responses to the combination therapy, thus supporting their predictive roles. We next evaluated the predictive performance of key signature genes of immune cell subsets showing high predictive potentials for the combination therapy, which included CD8-CXCL13, CD4-CXCL13 and pro-inflammatory macrophages of M ϕ -MMP9, as indicated by both the Pi and immune cell cluster enrichment analyses. We selected top-ranked signature genes, which were defined as marker genes with a score ≥ 0.1 measured using Logistic Regression (Table S2), in the signature gene lists of these immune cell subsets but not in those of other subsets in each major immune compartment. Of note, *CXCL9* and *CXCL11* were added to the core signature genes of pro-inflammatory macrophages considering their potential roles in T cell recruitment and ICB response, although they were not included by our selection procedure. We then evaluated the predictive performance of each gene using pre-treatment data of the combination therapy group. To be specific, we constructed scoring classifiers by the R package ROCR using the clinical outcomes of TNBC patients treated with the combination therapy, in which the clinical response of PR labeled as 1 while SD or PD labeled as 0, and the baseline fractions of immune cells expressing each gene in each major immune compartment. Signature genes with high AUC values could be explored as potential biomarkers for anti-PD-L1 immunotherapy.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis

Statistical analyses were performed as described in the Figure legends. Pearson correlation was used to estimate correlations among immune cell subsets. Survival was measured using the Kaplan-Meier method. Statistical significance was determined by Multivariate cox regression, Kruskal-Wallis test, Wilcoxon test, one-way ANOVA test and Limma moderated t-test with the Benjamini-Hochberg method for multiple comparisons correction.

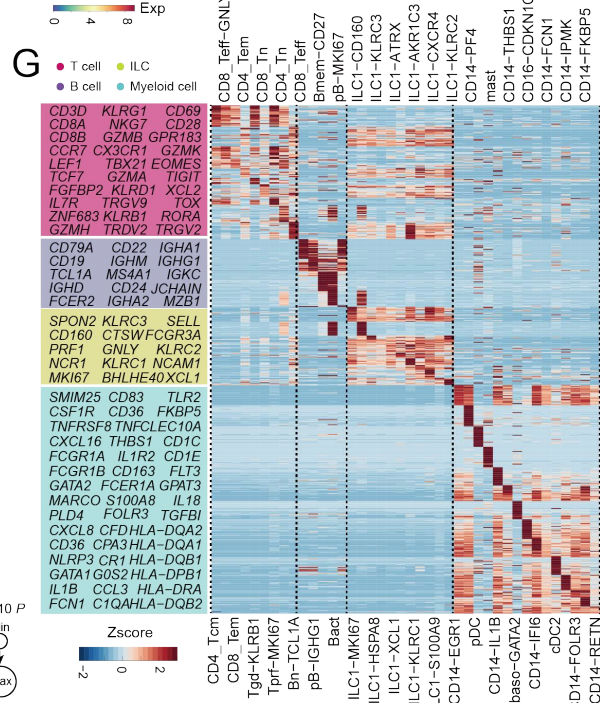
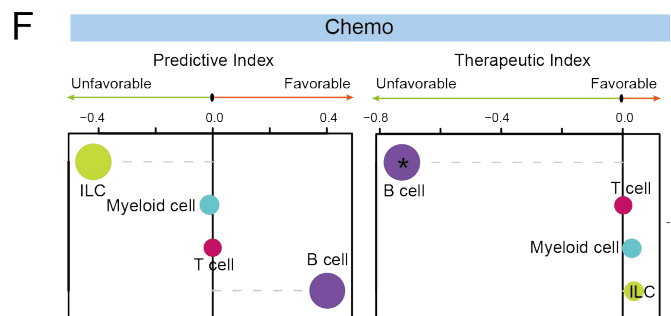
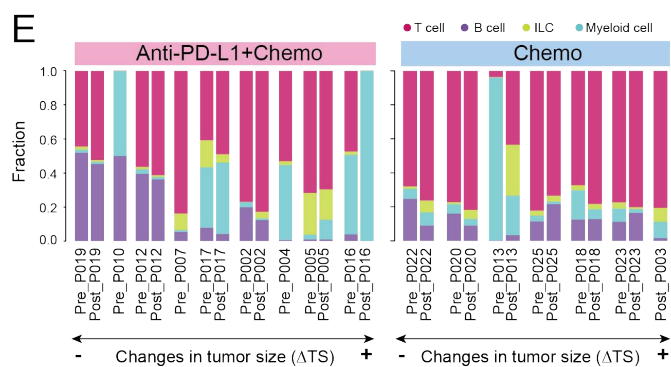
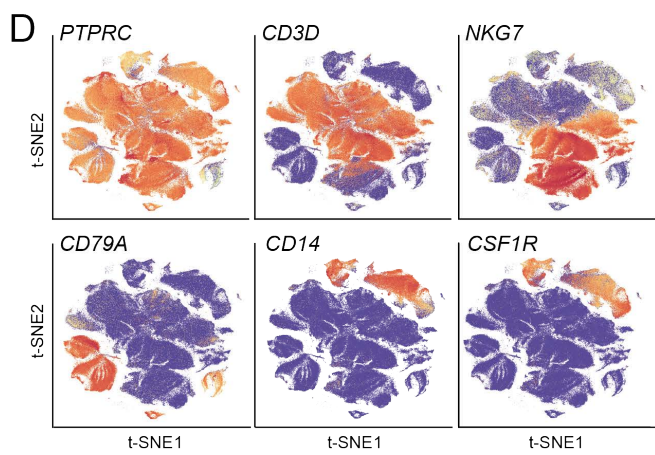
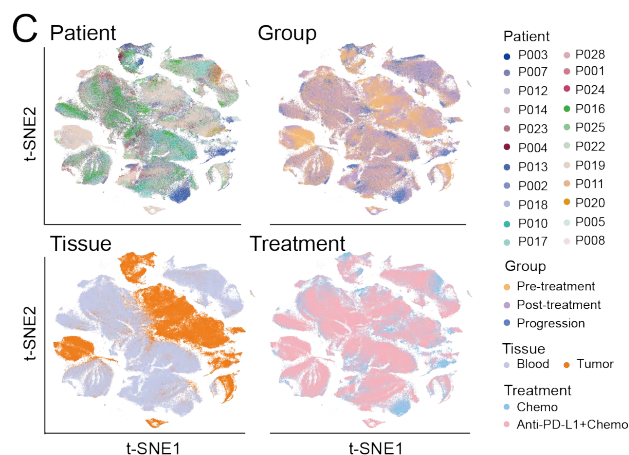
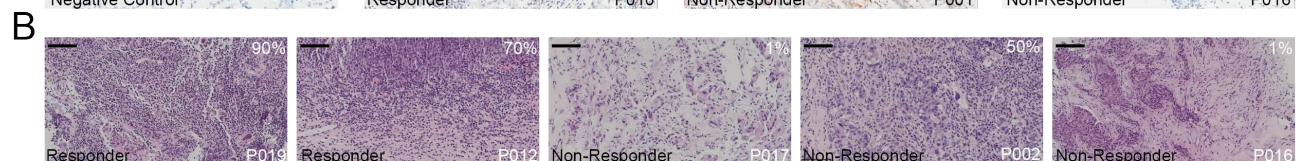
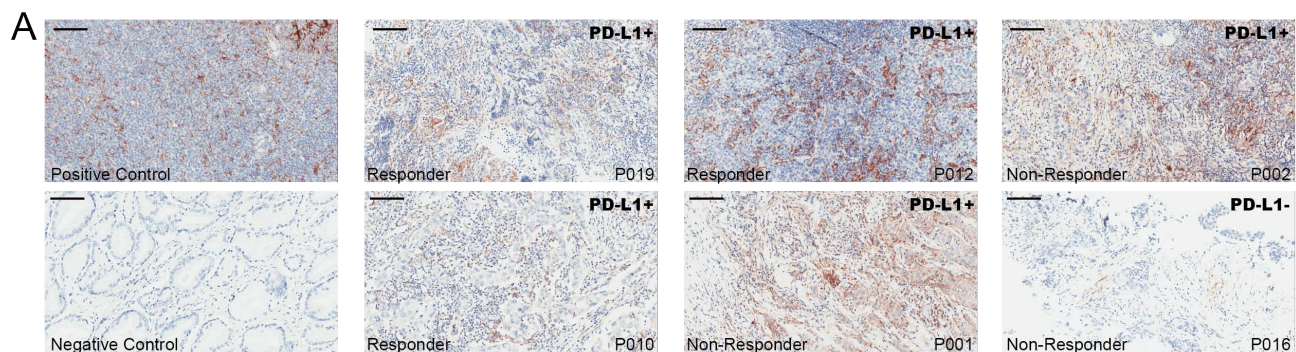
ADDITIONAL RESOURCES

Visualization and customized analyses of single-cell RNA-seq data of human patients with TNBC treated with paclitaxel or paclitaxel plus atezolizumab can be performed at http://TNBC_PD-L1.cancer-pku.cn.

Supplemental information

Single-cell analyses reveal key immune cell subsets associated with response to PD-L1 blockade in triple-negative breast cancer

Yuanyuan Zhang, Hongyan Chen, Hongnan Mo, Xueda Hu, Ranran Gao, Yahui Zhao, Baolin Liu, Lijuan Niu, Xiaoying Sun, Xiao Yu, Yong Wang, Qing Chang, Tongyang Gong, Xiuwen Guan, Ting Hu, Tianyi Qian, Binghe Xu, Fei Ma, Zemin Zhang, and Zhihua Liu



Supplementary Figure 1. The Immune Cell Atlas in TNBC Patients, related to Figure 1.

(A) IHC staining of PD-L1 protein expression in representative tumors. Positive control, tonsil; Negative control, prostate. Original magnification, x20; Scale bar, 100 μm .

(B) H&E staining in representative tumors. Original magnification, x20; Scale bar, 100 μm .

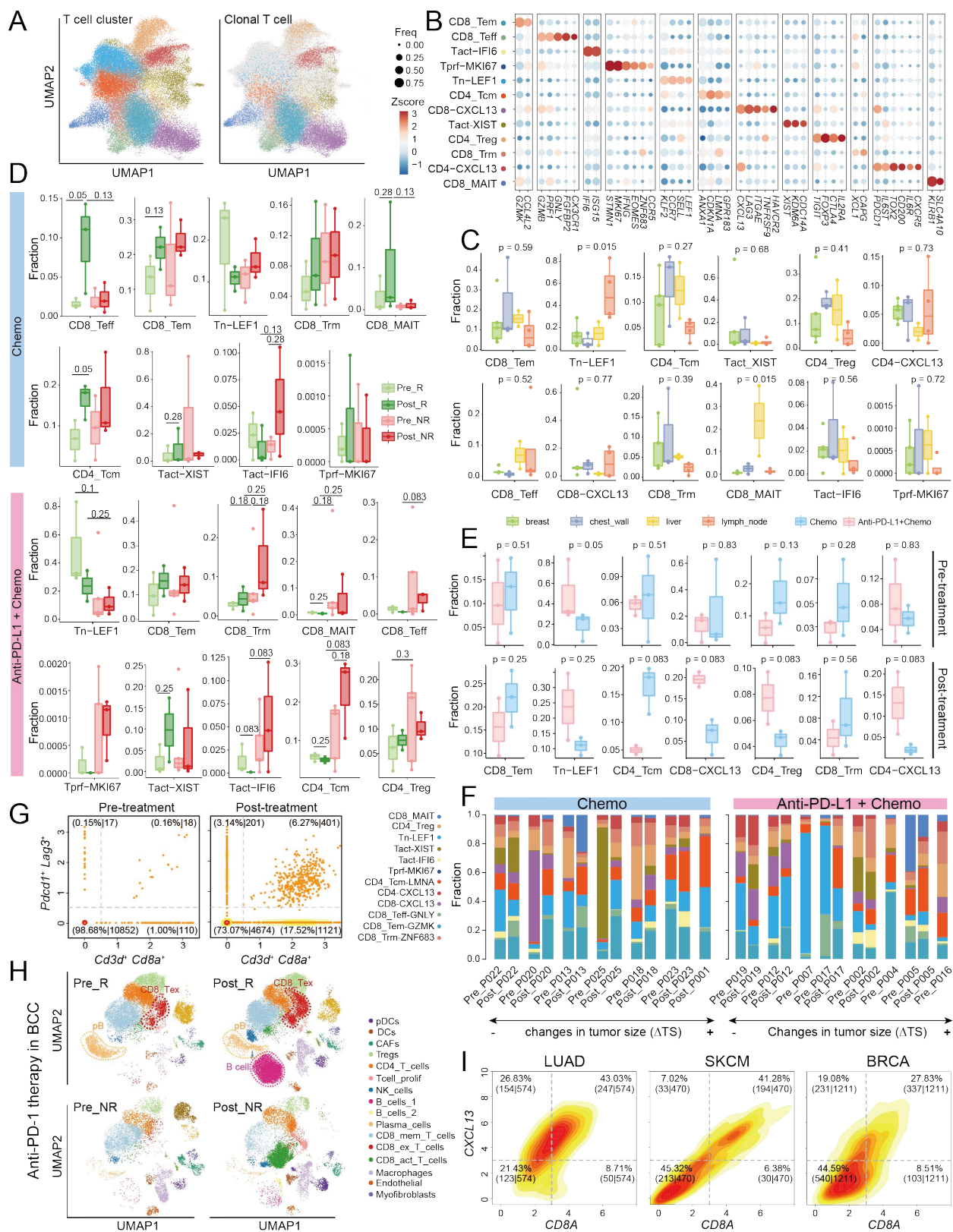
(C) t-SNE plots showing the origins of individual immune cells in TNBC patients.

(D) t-SNE plots showing the expressions of selected marker genes in major immune cell types.

(E) The cellular compositions of major immune cell types in TNBC tumors pre- and post-treatment.

(F) The predictive indices and therapeutic indices of major immune cell types in TNBC tumors treated with paclitaxel. Dot size represents the significance evaluated by $-\text{Log}_{10}(P\text{-value})$. *, $0.01 < P\text{-value} < 0.05$.

(G) The expression patterns of signature genes in distinct blood-derived immune cell clusters.



Supplementary Figure 2. Tumor-Infiltrating T Cell Subsets and Their Temporal Dynamics, related to Figure 2.

(A) UMAP plots showing T cell clusters and their clonal expansions in TNBC tumors. Clonal T cells were defined as those with TCR sequences harbored by at least two cells.

(B) The expression patterns of selected marker genes in distinct tumor-infiltrating T cell clusters.

(C) Boxplots showing the tissue origins of distinct tumor-infiltrating T cell clusters. One-way ANOVA test.

(D) Boxplots showing the temporal dynamics of distinct tumor-infiltrating T cell clusters. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

(E) Boxplots showing the cellular compositions of distinct T cell clusters in responsive tumors treated with different regimens. Kruskal-Wallis test.

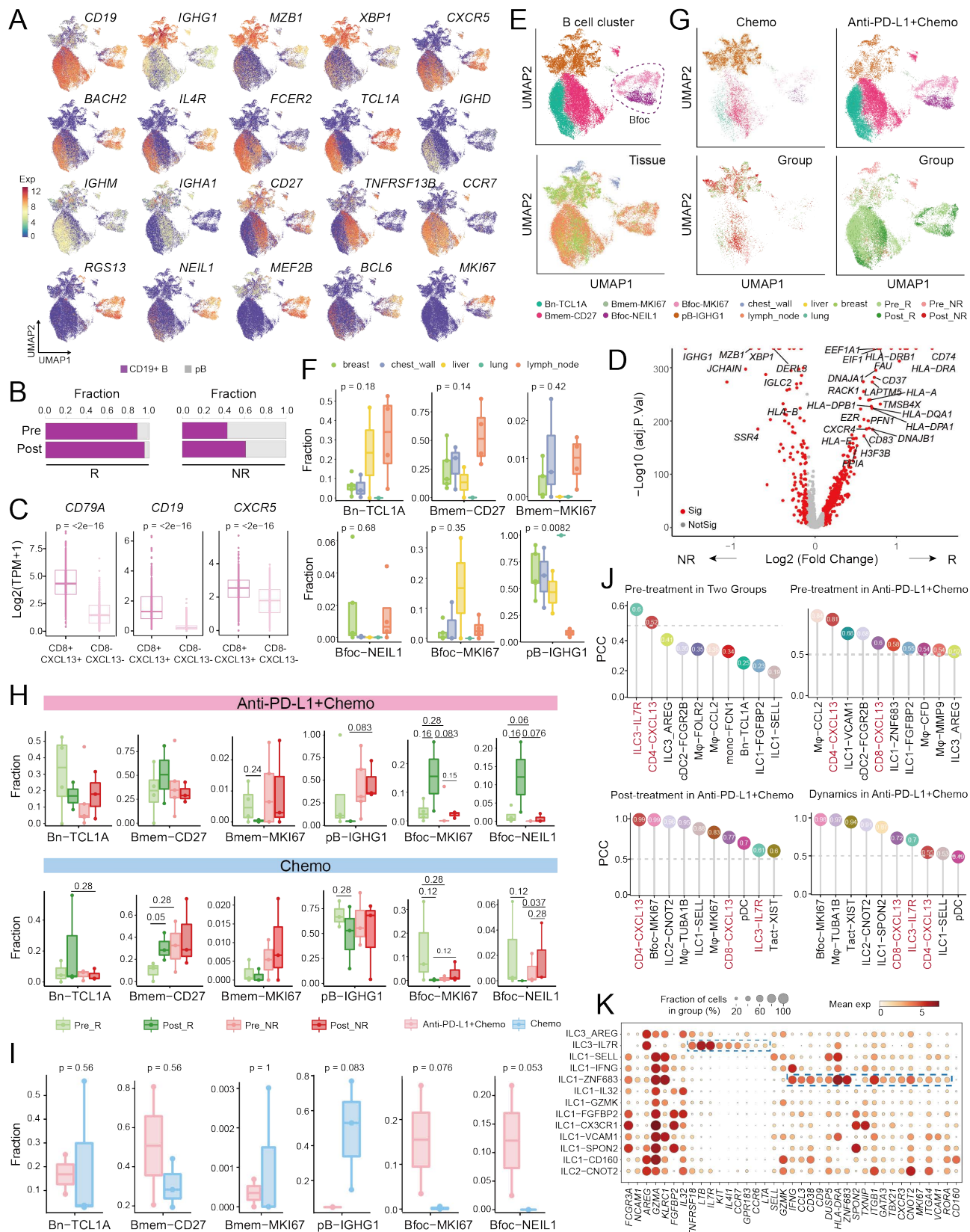
(F) Barplots showing the cellular compositions of distinct tumor-infiltrating T cell clusters in individual TNBC patients. Patients were ranked based on their tumor size changes.

(G) *In silico* flow cytometry (FACS) plots showing the alteration of $Cd3d^+ Cd8a^+ Pdcd1^+ Lag3^+$ T cells in T11-Apobec murine model of BC following ICB treatment (Hollern et al., 2019). The geometric mean of expression levels of *Cd3d* and *Cd8a*, and those of *Pdcd1* and *Lag3* were used to evaluate the percentages of $Cd3d^+ Cd8a^+ Pdcd1^+ Lag3^+$ T cells.

(H) UMAP plots showing the alterations of tumor-infiltrating immune cell subtypes in BCC patients pre- and post-treatment with ICB (Yost et al., 2019).

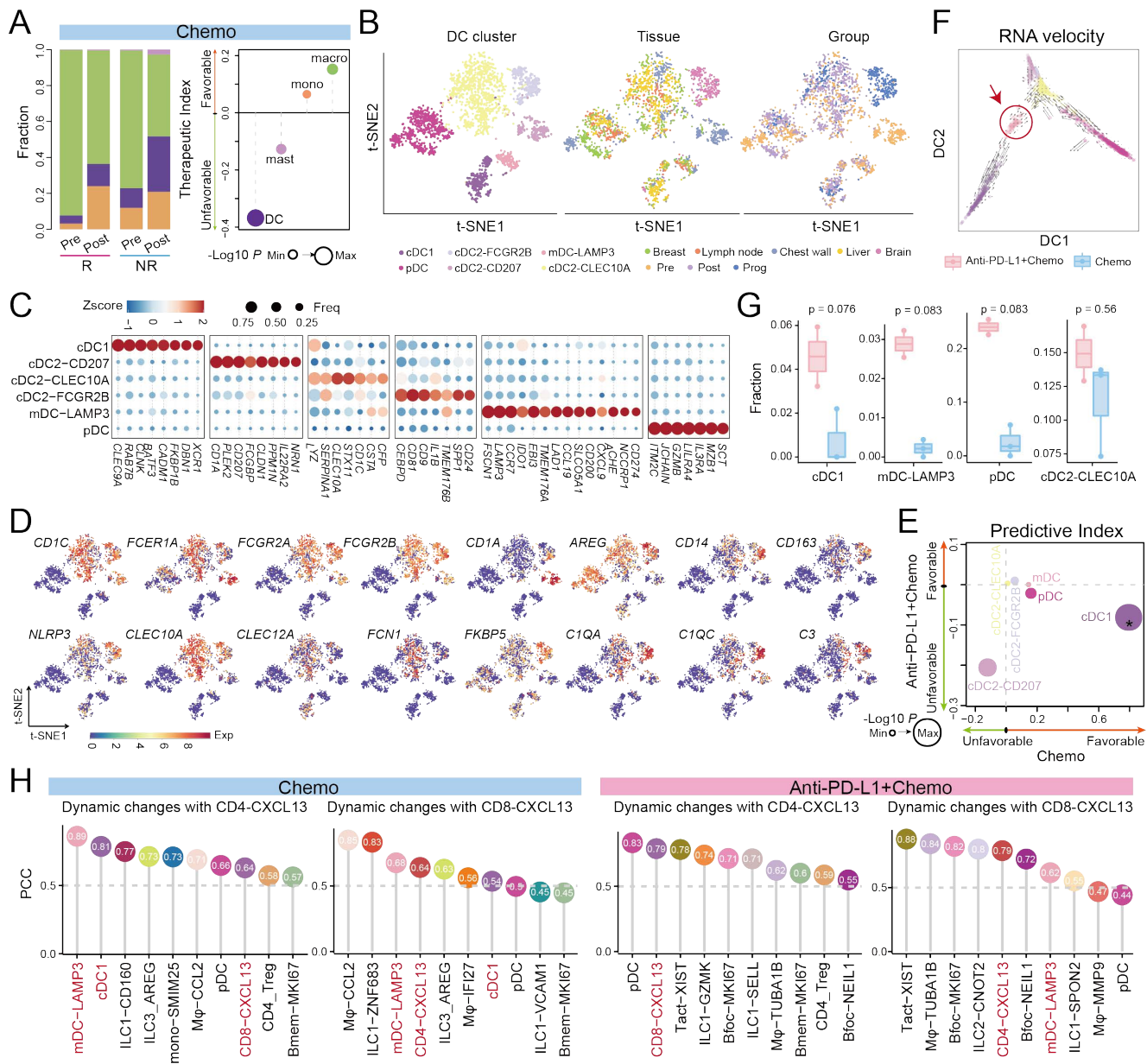
(I) Density plots showing the proportions of tumor samples that express high levels of *CD8A* and *CXCL13* in TCGA lung adenocarcinoma (LUAD), skin cutaneous melanoma (SKCM) and breast invasive carcinoma (BRCA). Expression level was measured by Log2 (TPM+1).

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5× the interquartile range.



Supplementary Figure 3. Features of Tumor-infiltrating B Cell Clusters, related to Figure 3.

- (A) UMAP plots showing the expression patterns of selected genes in tumor-infiltrating B cells.
 - (B) Barplots showing the compositional changes of *CD19*⁺ B and pB cells in responders (R) and non-responders (NR) of the combination therapy group.
 - (C) Boxplots showing the expression levels of selected genes in TCGA BRCA patients grouped by *CD8A* and *CXCL13* expressions in Figure S2I.
 - (D) The volcano plot showing DEGs in B cells enriched in pre-treatment responsive and non-responsive tumors of the combination therapy group. Sig, significant; BH adjusted *P*-value < 0.01 and Log2 (fold change) ≥ 0.1 or Log2 (fold change) ≤ -0.1. Limma moderated t-test.
 - (E) UMAP plots showing the distributions and tissue origins of tumor-infiltrating B cell clusters.
 - (F) Boxplots showing the tissue origins of distinct B cell clusters. One-way ANOVA test.
 - (G) UMAP plots showing the distributions of B cell clusters in different treatment groups.
 - (H) Boxplots showing the temporal dynamics of distinct B cell clusters. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.
 - (I) Boxplots showing the cellular compositions of distinct B cell clusters in responsive patients following different treatment regimens. Kruskal-Wallis test.
 - (J) Correlations of distinct tumor-infiltrating immune cells with Bfoc in different treatment groups. PCC, Pearson correlation coefficient.
 - (K) The expression patterns of selected marker genes in distinct tumor-infiltrating ILC clusters.
- For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5 × the interquartile range.



Supplementary Figure 4. Characteristics of Tumor-infiltrating DC Subsets, related to Figure 4.

(A) The cellular compositions and dynamics of major myeloid subpopulations in TNBC tumors treated with paclitaxel.

(B) t-SNE plots showing the distributions of DC subsets.

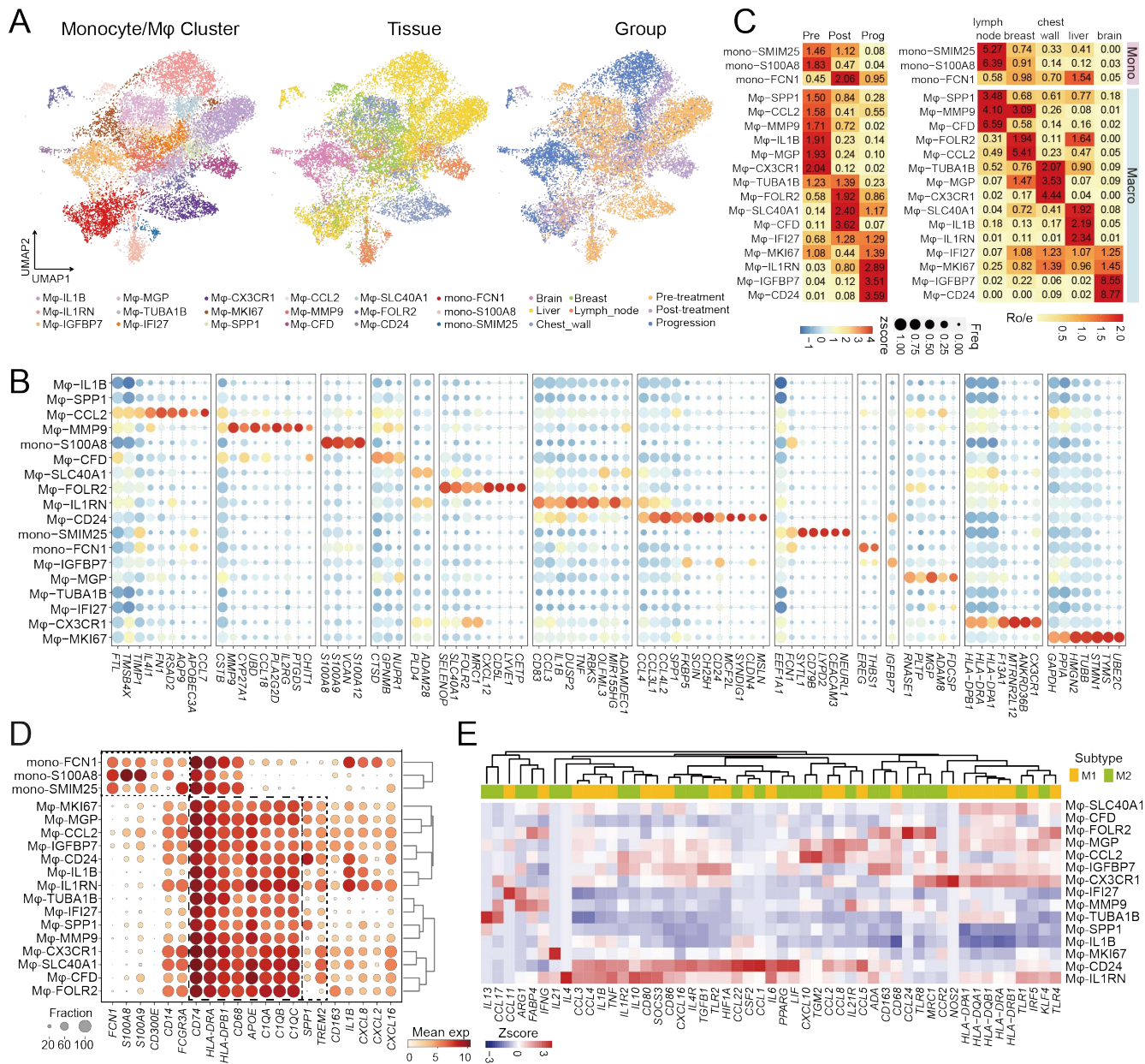
(C) Bubble heatmap and (D) t-SNE plots showing the expression patterns of selected marker genes in distinct DC clusters.

(E) The predictive indices of distinct DC subsets. Dot size represents the geometric mean of $-\text{Log}_{10}(P\text{-value})$ in different treatment groups. * bottom, $0.01 < P\text{-value} < 0.05$ for chemotherapy.

(F) The developmental trajectory of distinct DC clusters inferred by RNA velocity. DC, diffusion component.

(G) Boxplots showing the cellular compositions of distinct DC clusters in responsive patients following different treatment regimens. Box middle lines, median; box limits, upper and lower quartiles; box whiskers, $1.5 \times$ the interquartile range. Kruskal-Wallis test.

(H) Correlations of distinct tumor-infiltrating immune cell subsets with CD4-CXCL13 and CD8-CXCL13 in their cellular proportion dynamics in different treatment groups. PCC, Pearson correlation coefficient.



Supplementary Figure 5. The Heterogeneity of Monocytes/Macrophages in TNBC Tumors, related to Figure 5.

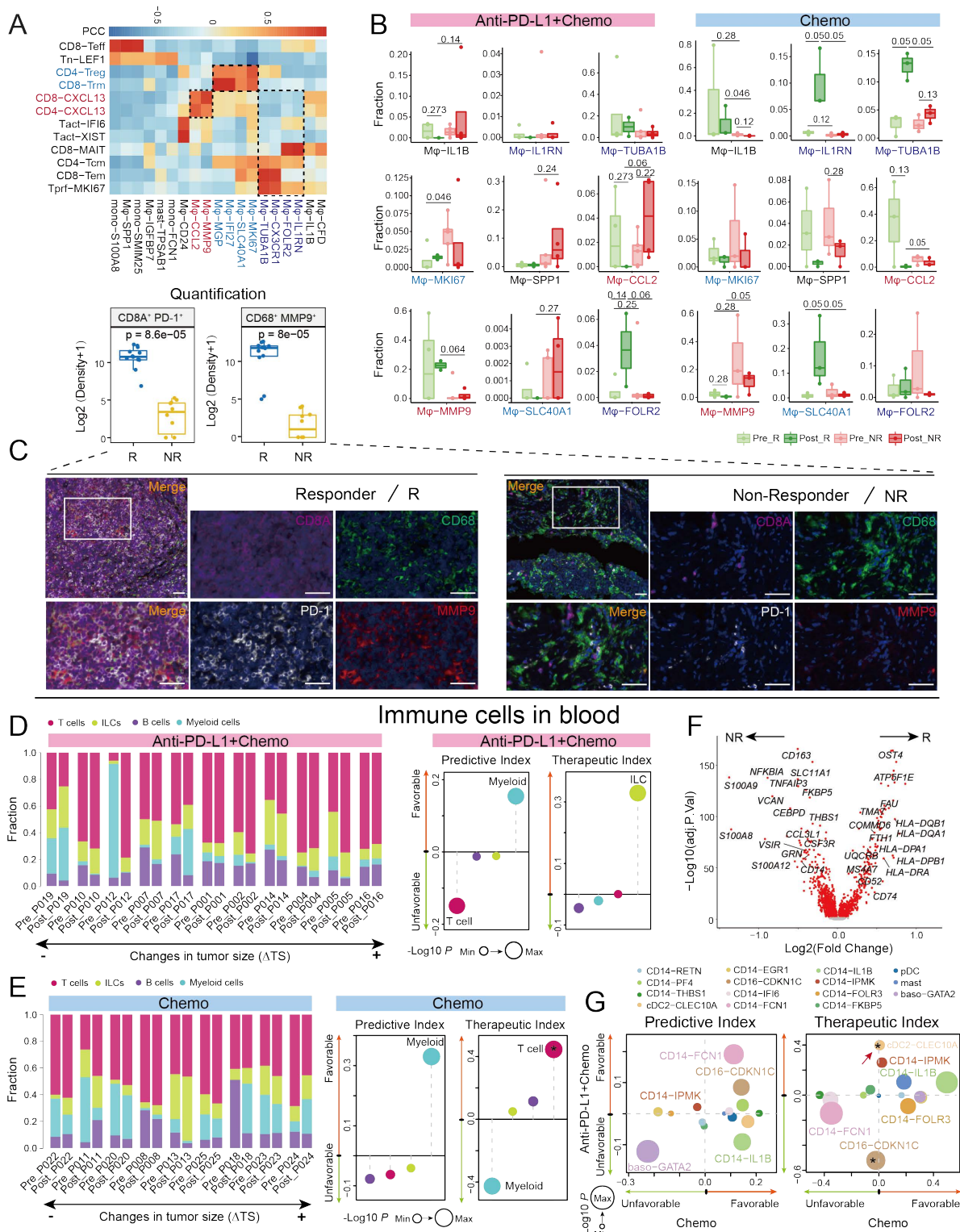
(A) UMAP plots showing the monocyte/macrophage cell clusters in TNBC tumors and their distributions in different tissues and groups.

(B) The expression patterns of signature genes of distinct monocyte/macrophage cell clusters.

(C) The enrichment of distinct monocyte/macrophage clusters in different groups and tissues (STAR Methods).

(D) Bubble heatmap showing the expressions of selected genes.

(E) Heatmap showing the expressions of canonical M1 or M2 phenotype-related genes.



Supplementary Figure 6. Features and Dynamics of Monocycles/Macrophages in TNBC Patients, related to Figure 5.

(A) Heatmap showing correlations of distinct monocycle/macrophage clusters with T cell clusters in their baseline cellular proportions in TNBC patients treated with paclitaxel plus atezolizumab. PCC, Pearson correlation coefficient.

(B) Boxplots showing the temporal dynamics of distinct macrophage subsets in TNBC tumors. Black lines with different lengths indicate which two groups were compared. Kruskal-Wallis test.

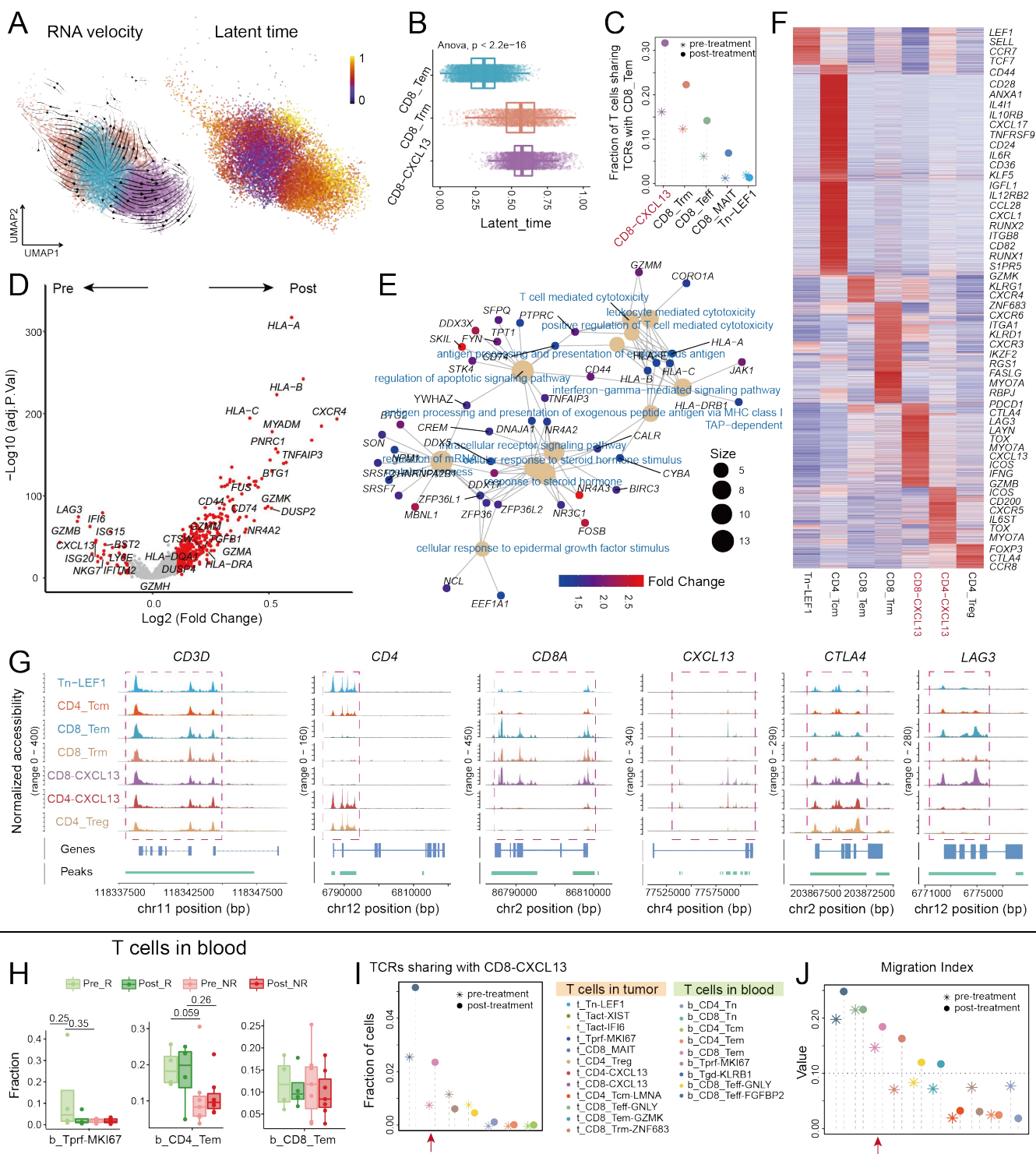
(C) Multi-color IHC staining in one representative responder (P019) and one non-responder (P016) of the combination therapy group. Scale bar, 30 μ m. Each dot in boxplot represents one view. Wilcoxon test.

(D) The cellular compositions, predictive indices and therapeutic indices of blood-derived immune cells in TNBC patients treated with the combination therapy or (E) the chemotherapy. *, $0.01 < P\text{-value} < 0.05$.

(F) The volcano plot showing DEGs of baseline blood monocytes in responders and non-responders of the combination therapy group. BH adjusted $P\text{-value} < 0.01$. Limma moderated t-test.

(G) The predictive indices and therapeutic indices of distinct blood-derived myeloid subsets. * left, $0.01 < P\text{-value} < 0.05$ for the combination therapy group.

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, $1.5 \times$ the interquartile range.



Supplementary Figure 7. Transcriptomic and Epigenetic Features of CD8-CXCL13 T cells, related to Figure 6.

(A) The developmental trajectory of CD8⁺ T cell subsets of Tem, Trm and CD8-CXCL13 inferred by RNA velocity.

(B) Boxplots showing the inferred latent times of CD8⁺ T cell clusters based on their developmental trajectory.

(C) The percentages of clonal T cells that shared TCRs with CD8 Tem in distinct T cell subsets in responsive tumors pre- and post-treatment with paclitaxel plus atezolizumab.

(D) The volcano plot showing DEGs of CD8-CXCL13 in responsive patients pre- and post-treatment with the combination therapy. BH adjusted *P*-value < 0.01. Limma moderated t-test.

(E) The enriched pathways for genes highly expressed in CD8-CXCL13 in responsive patients following the combination therapy.

(F) Heatmap showing the signature peaks and associated genes in distinct T cell clusters.

(G) ATAC-seq tracks showing the chromatin accessibilities of *CD3D*, *CD4*, *CD8A*, *CXCL13*, *CTLA4* and *LAG3* in distinct tumor-infiltrating T cell clusters.

(H) Boxplots showing the temporal dynamics of selected blood-derived T cell subsets in TNBC patients treated with the combination therapy.

(I) The percentages of clonal T cells that shared TCRs with tumor CD8-CXCL13 in distinct T cell subsets in responsive patients pre- and post-treatment with the combination therapy.

(J) The STARTRAC migration index of blood CD8 Tem cells in responsive patients pre- and post-treatment with the combination therapy.

For all box plots: box middle lines, median; box limits, upper and lower quartiles; box whiskers, 1.5× the interquartile range.